

assignments in

Junior Science

book 7

Genetics

CHRIS GREEF
SUADA BILALI



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resources



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Suada Bilali completed her Bachelor of Science and Dip.Ed. at Macquarie University. She has spent the majority of her teaching career at inner west schools in Sydney, teaching students from diverse backgrounds. Suada has written a series of books for Blake Education and has also worked as a Science consultant and written a series of junior text books for Longman.

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An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
www.blake.com.au



ISBN 978 1 92209 082 9

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Each book in the *Assignments in Junior Science* series provides a selection of work sheets covering one or two major topics. The work sheets are generally self-contained, supplying the information necessary for students to formulate answers. As students work through these activities, they will interpret and analyse data including text, tables, graphs and diagrams; and organise and represent data in a variety of formats. They will also utilise basic scientific skills such as working with formulas, completing calculations, making generalisations and explaining what they have discovered. The activities incorporate a range of theoretical and practical scientific skills, and also involve the development of literacy skills.

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Meiosis is a special type of cell division that occurs in the sex organs of animals and plants; for example, in the ovaries and testes of animals. Meiosis results in the production of *ova*, which are eggs (singular: *ovum*), in females and sperm in males. Ova and sperm are also called gametes. Haploid gametes are produced from diploid cells.

Meiosis requires two cell divisions which result in four daughter cells. Each daughter cell has half the number of chromosomes as the parent cell (daughter cells are haploid, parent cell is diploid), therefore they are not identical to the parent cell nor to each other. Meiosis is often called the **reduction division**.

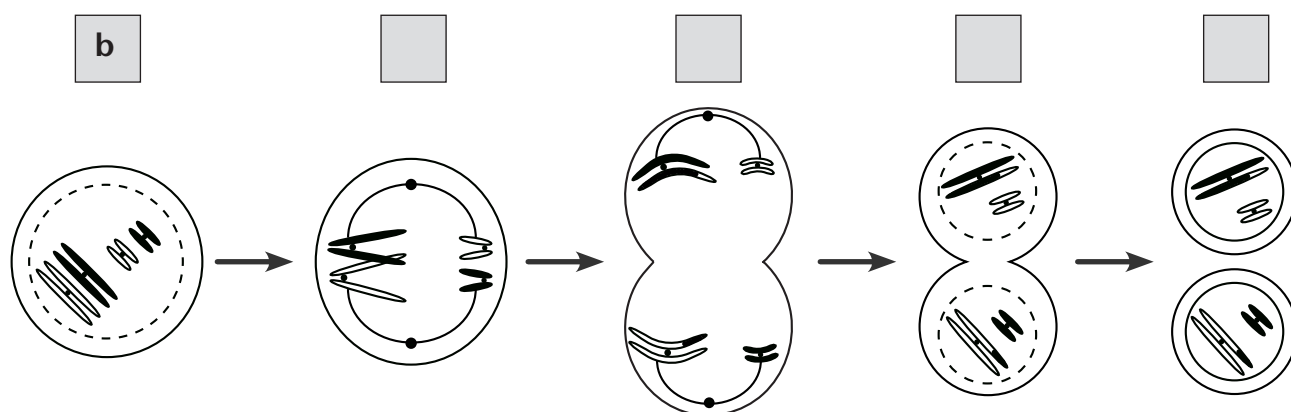
Meiosis – stage 1

1 The text below describes what is occurring in each part of the following diagram.

- Use the word bank to complete the descriptions.
 - Then match each description to the correct part of the diagram by writing the appropriate letter in the grey box. One has been done for you.
- a** Two new cells have formed. Each cell contains only one _____ from each homologous pair.
- b** Spindle begins to appear. _____ membrane begins to disappear. Chromosome doubles pair up with another set of chromosome doubles of the same size. These are called _____ chromosomes.
- c** Chromosomes are now completely _____. Nuclear membrane begins to form around the chromosomes. The _____ begins to disappear. The cell begins to separate.
- d** The homologous pair of chromosomes begins to move _____, to the opposite sides of the _____.
- e** The homologous chromosomes line up at the centre of the _____. Each _____ of chromosomes is attached to the spindle.

Word bank

- spindle
- chromosome
- Nuclear
- pair
- separated
- cell
- cell
- apart
- homologous



Meiosis – stage 2

2 The text below describes what is occurring in each part of the diagram on the next page.

- Use the word bank to complete the descriptions.
 - Then match each description to the correct part of the diagram by writing the appropriate letter in the grey box.
- a** Cell _____ is complete. Four new cells have formed. Each cell carries half the number of chromosomes of the _____ cell.
- b** Chromosomes begin to _____ and move apart.
- c** At the beginning of stage 2, the _____ reappears and the nuclear _____ begins to disappear.

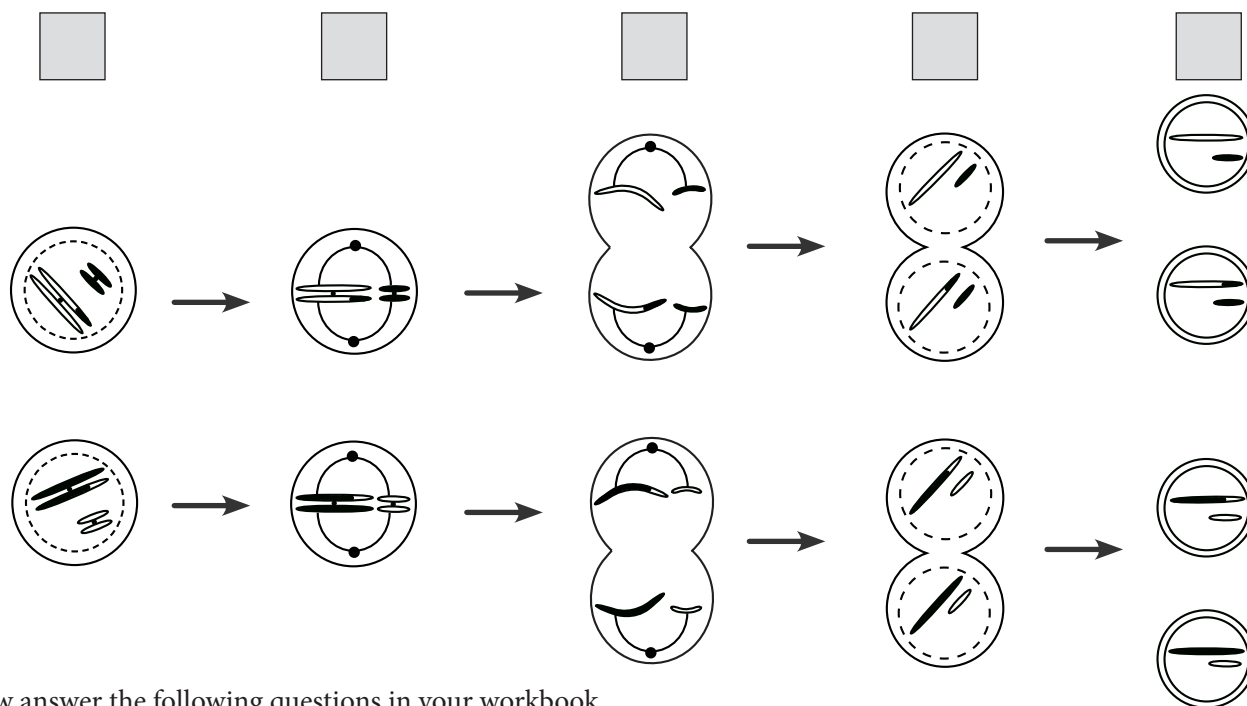
Word bank

- disappear
- spindle
- division
- parent
- chromosome
- split
- separate
- cell
- membrane



d Chromosomes move towards the poles and are completely separated. The spindle begins to _____. Nuclear membrane begins to form. The cell begins to _____.

e Chromosomes line up at the equator of the _____. Each _____ is attached to the spindle.



Now answer the following questions in your workbook.

3 Describe the differences between stage 1 and stage 2 meiosis.

4 Define each term.

a Meiosis

d Poles

b Homologous chromosomes

e Haploid

c Equator

f Diploid

5 What do you think would happen at fertilisation if the chromosomes in the sex cells didn't halve in number during meiosis?

6 Explain why **crossing over** is so important. *Hint:* Crossing over of chromosomes occurs in the second diagram.

7 a In stage 2, how many sex cells were produced?

b Are they all similar?

c Why is this significant?

8 **Modelling meiosis**

- Your teacher will supply you with 2 different coloured pieces of plasticine or modelling clay.
- Construct two large pairs of chromosomes, one pair of each colour, and two small pairs of chromosomes, one pair of each colour, as is shown in the first part of the Stage 1 Meiosis diagram on the previous page.
- Use your plasticine chromosomes to reconstruct meiosis as shown in the Meiosis stage 1 and stage 2 diagrams. Move your model chromosomes to show what is occurring in each diagram, until you reach the last diagram.

a Did you end up with 4 different gametes at the end of the activity?

b What helped to increase the differences in the chromosomes?

c What was useful about using this model? For example, did it help you understand meiosis?

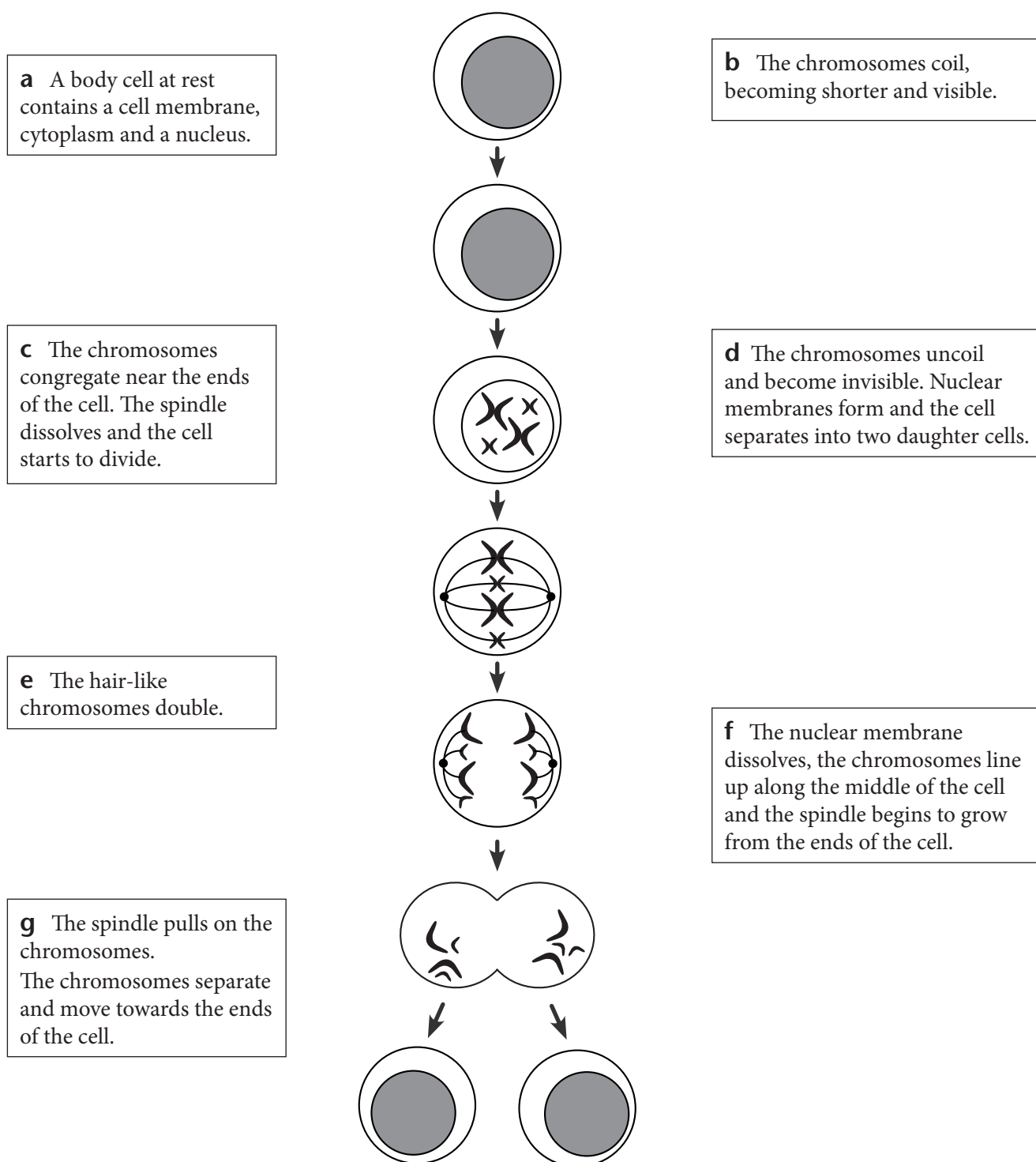
d What were the limitations of using this model?



Cells divide in order to make more cells. Usually the new cells (or daughter cells) are the same as the original ones (the parent cells). This type of cell division is called **mitosis**. Mitosis occurs when organisms grow bigger, when organisms replace lost or damaged cells, and when simple organisms reproduce asexually. Sometimes parent cells divide to develop into sex cells (or gametes); these egg and sperm contain half the normal number of chromosomes and join later during fertilisation to produce the first cell of a new offspring. This type of cell division is called **meiosis**.

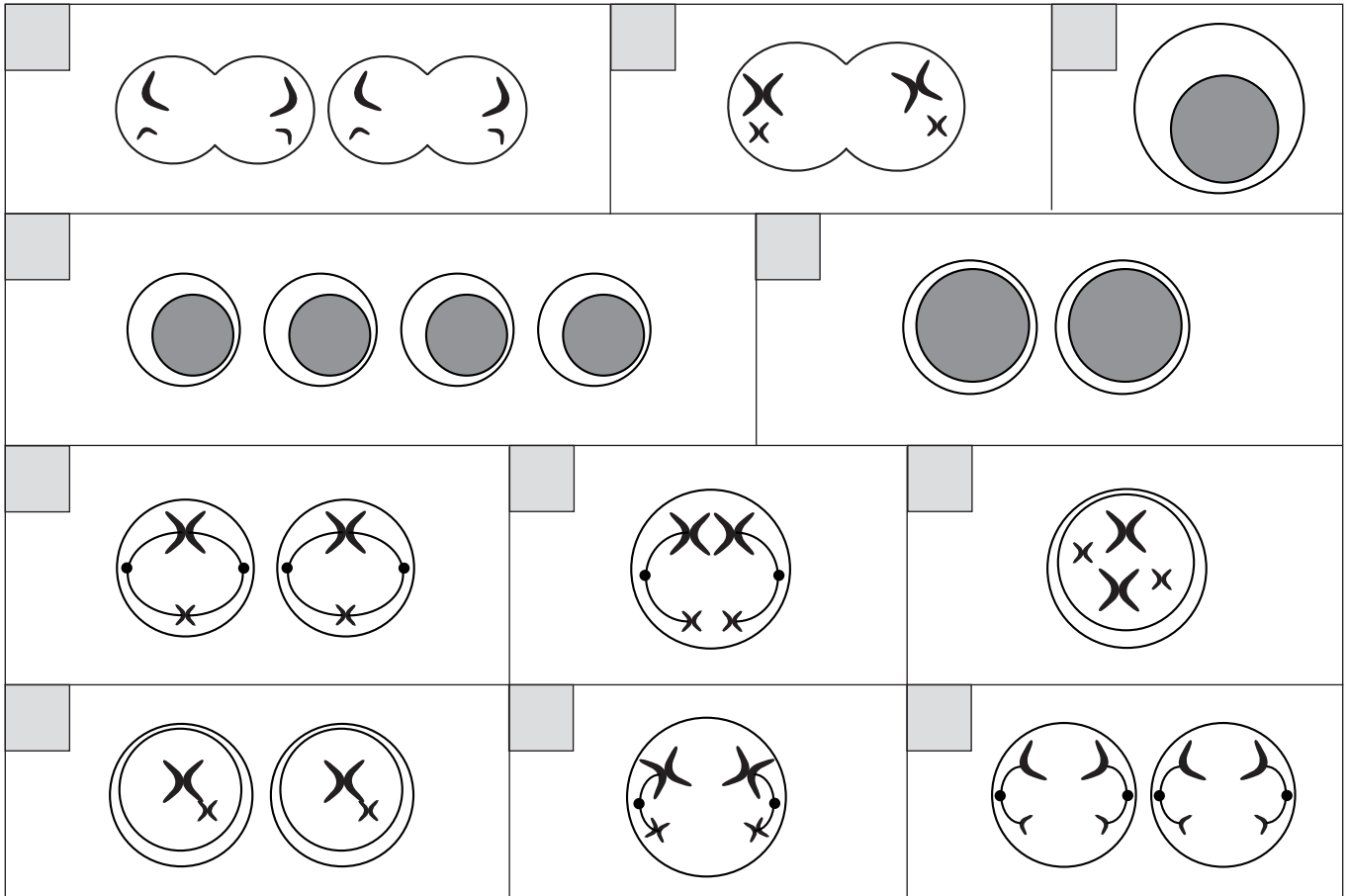
When we refer to identical daughter cells, we mean that the chromosomes, which are the carriers of genetic information, are **exactly the same**. The cells themselves are smaller and similar to each other.

- 1** The diagram below shows a cell undergoing mitosis. Draw lines to match each description to the correct part of the diagram.





- 2** Use the information in question 1 to write in your workbook an explanation for how mitosis occurs. Remember to write a brief introduction in which you define mitosis. Then write the explanation sequence using conjunctions such as *when*, *after* and *because*. Finally, write a conclusion where you state the purpose of mitosis.
- 3** The diagrams below show different stages of a cell undergoing meiosis. Write the letters **a** to **k** in the grey boxes to show the sequence in which meiosis occurs.



- 4** In your workbook write short descriptive statements for each stage of meiosis shown above. Use the descriptive statements about mitosis on the previous pages as a model.
- 5** Referring to the descriptive statements you completed for question 4, write in your workbook an explanation for how meiosis occurs.
- 6** Complete the table below to outline the differences between meiosis and mitosis.

	Mitosis	Meiosis
Where does it occur?		
How many stages of division?		
How many cells result at the end of the division?		
In humans, how many chromosomes are there at the end of the division?		
Are the daughter cells identical to the parent cell?		
Are the daughter cells identical to each other?		



Chromosomes with their genes are made up of the chemical **DNA** or **deoxyribonucleic acid**. DNA is a long molecule that stores all genetic information. It looks like a double helix or a twisted ladder, as you can see in the diagram at the right.

DNA has only four repeated subunits, the **nucleotides**. In humans, a DNA molecule is about 2 cm long. In a 1 mm section of a DNA molecule there can be 8 million nucleotides. Each nucleotide is made up of a sugar unit, a phosphate unit and a base unit. The phosphate links the sugars to make the long strand (i.e. sugar – phosphate – sugar – phosphate – sugar). The bases of one strand lock onto the bases of the other strand to form the double strand.

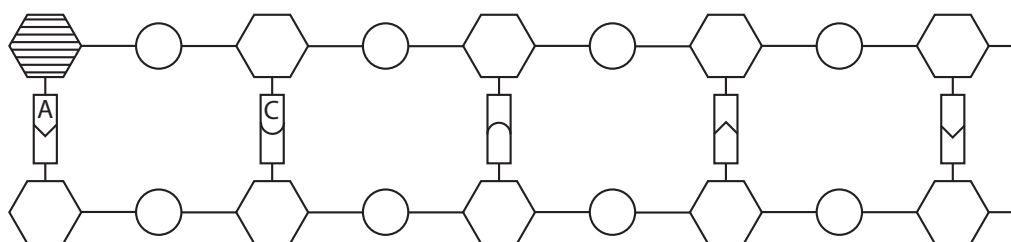
Although the sugar and phosphate units are all the same, there are four kinds of bases: adenosine (A), cytosine (C), guanine (G) and thymine (T). The bases only join in certain pairs: A – T and C – G. In order to code for a particular gene such as tongue rolling, the DNA section of a particular chromosome is a certain length and the bases are in a particular order.



1 Identify the chemicals sugar, phosphate and the four bases of the DNA section by:

- shading **sugar** horizontally
- shading **phosphate** vertically
- labelling each base as A, C, G or T.

One sugar and two bases have been identified for you.

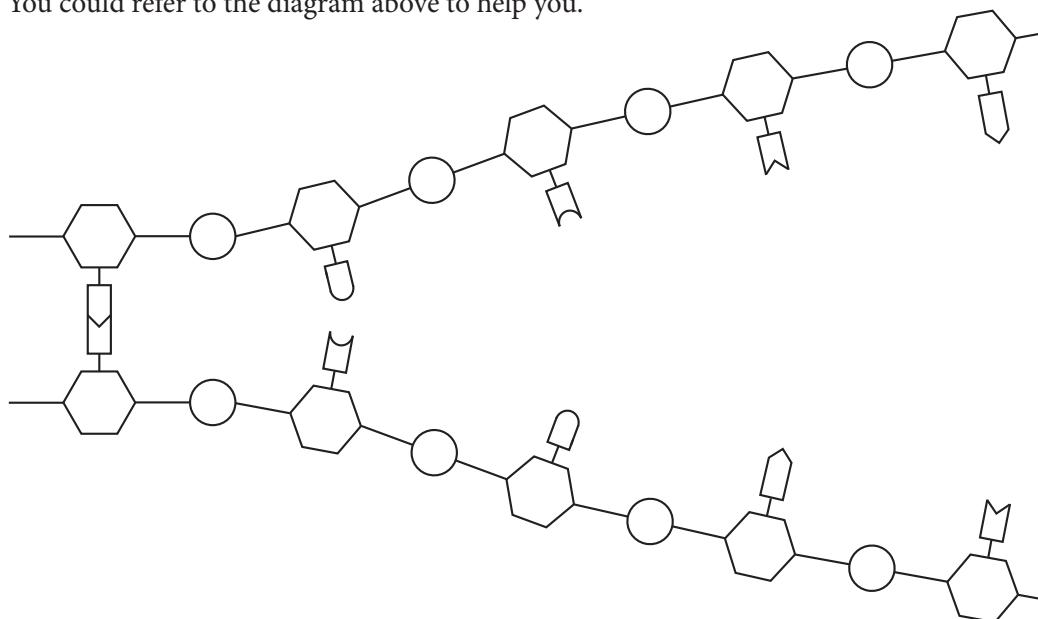


When a cell divides, it first has to make a copy of its DNA molecules. First, the double helix untwists. Then it splits open into two strands. The nucleotides which are floating around in the nucleus come into contact with the open strands and matching bases lock together. After a few minutes, two new double strands have been produced. It is important that the two new double strands are identical to the original.

2 The diagram below shows a DNA molecule splitting into two strands. For the parts that have already split:

- draw the nucleotides that will lock together with the open strands
- draw sugar and phosphate units to join the nucleotides you drew.

You could refer to the diagram above to help you.



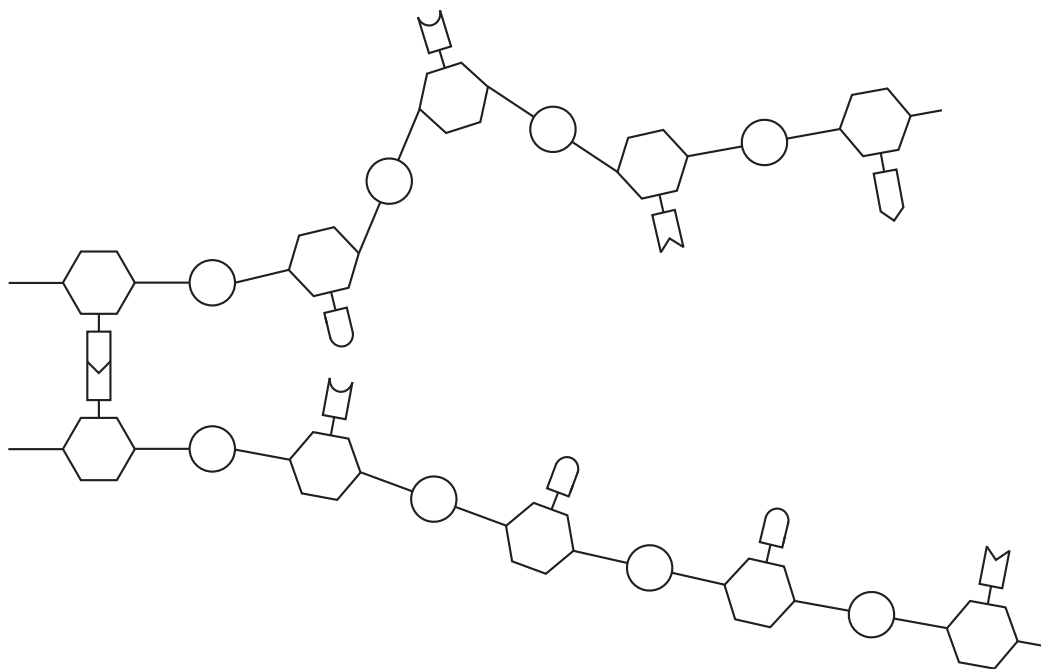


3 Compare the new double strands with the original double strand. Are there any differences?

4 Compare the new double strands with each other. Are there any differences?

5 Explain how DNA duplication reduces the likelihood of errors.

Sometimes things go wrong during DNA duplication. In the diagram below, two mistakes have occurred.



6 As you did for question two, draw the double strands for the open sections of the diagram above. There will be two mistakes in the new double strands.

7 Describe the two mistakes in the new double strands.

Sometimes DNA can rectify small mistakes like those above. At other times, mistakes result in a new gene which creates a new characteristic in the organism. For example, it may no longer produce a certain enzyme which is needed during respiration, but produces a different enzyme instead. As a result, the cell will no longer be able to respire.

8 Explain what would happen to a cell if it could no longer respire.

Many DNA changes or mutations are harmless, such as a new gene that codes for a different eye colour. Some mutations are fatal and a few are beneficial. Mutations can be caused by radiation, certain chemicals and unusual heat.

9 Define *mutation*.

10 Explain why a pregnant woman should not have an X-ray.

11 If a pregnant woman takes medication without consulting her doctor, how might this affect the growing foetus?



1 Use the word bank to complete the information about DNA.

Word bank

- | | | | | |
|-----------|-------------|---------------|---------------|-----------|
| • differs | • important | • helix | • nucleotides | • organic |
| • code | • together | • significant | • things | • egg |
| | • strand | • discovery | • zygote | |

Your life began when one _____ and one sperm united to form a _____ – from a single cell to you! How did this single cell turn into thousands and thousands of different cells with specialised functions? Where did the instructions for this come from?

The answer lies with DNA. DNA is short for deoxyribonucleic acid. It is an _____ molecule that contains all the genetic information needed to make you, and all other living things. The DNA of all living _____ is unique.

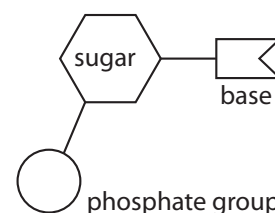
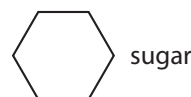
The _____ of DNA was led by James Watson and Francis Crick in 1953, with the help of Rosalind Franklin and Maurice Wilkins. It was one of the most famous and _____ discoveries in Science.

The basic unit of DNA is the nucleotide. (See Fig 1a.) DNA is made up of millions of _____ linked together like a chain. A nucleotide consists of a sugar, a phosphate and a nitrogenous base.

There are four different nitrogenous bases in DNA: adenine (A), thymine (T), cytosine (C) and guanine (G). The bases pair up, always A – T and G – C. (See Fig 1b.) The order of these bases is what determines the chemical code. One _____

_____ of nucleotides is joined together to another strand of nucleotides with the help of the bases. The two strands of nucleotides spiral with each other to make a shape called the _____. DNA is often referred to as a double helix because of the two strands of nucleotides (see Fig 2). There is another molecule like DNA in every cell. It is called RNA (ribonucleic acid). RNA is a single strand and contains the sugar ribose, whereas DNA has deoxyribose. Another difference is that RNA does not have the base thymine; instead it has uracil, symbol U.

The order of the bases in DNA is very _____, as this codes for protein. Proteins are long chains of amino acids joined _____ like beads on a chain. In 1960, 20 different amino acids were identified. It was discovered that three bases of m-RNA code for one amino acid (see Table 1).

Nucleotide**Figure 1a****DNA parts**

sugar



phosphate group

Four bases

Guanine



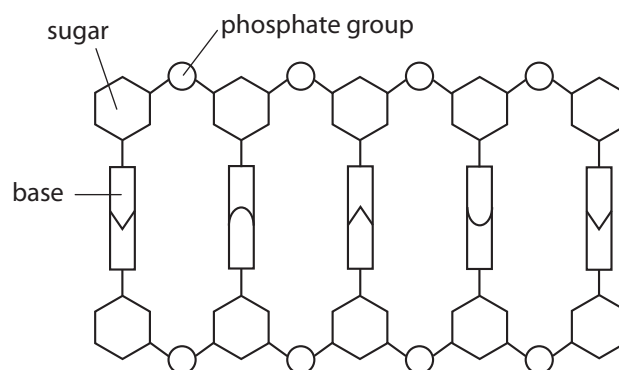
Cytosine



Adenine



Thymine

Figure 1b**Two strands of nucleotides: DNA****Figure 2**



All the codes were organised into a table (see Table 1) which shows the three bases (each abbreviated by a capital letter) in front of an amino acid (abbreviated to three letters). The full names of the amino acids are listed in Table 2; 'stop' indicates the end of an amino acid sequence – the protein is complete.

Remember, DNA is **opposite** to the _____ of m-RNA. That is, m-RNA is the complementary copy of DNA. But it _____ in one base: instead of **thymine**, RNA has **uracil**.

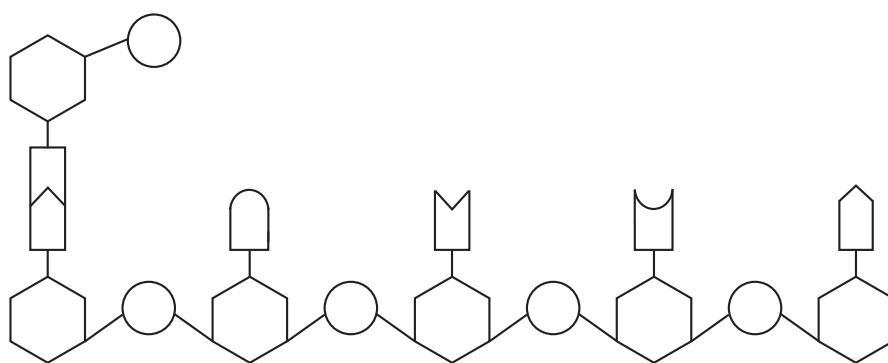
Table 1

m-RNA amino acid triplet codes			
UUU Phe	UCU Ser	UAU Tyr	UGU Cys
UUC Phe	UCC Ser	UAC Tyr	UGC Cys
UUA Leu	UCA Ser	UAA stop	UGA stop
UUG Leu	UCG Ser	UAG stop	UGG Try
CUU Leu	CCU Pro	CAU His	CGU Arg
CUC Leu	CCC Pro	CAC His	CGC Arg
CUA Leu	CCA Pro	CAA Gln	CGA Arg
CUG Leu	CCG Pro	CAG Gln	CGG Arg
AUU Ile	ACU Thr	AAU Asn	AGU Ser
AUC Ile	ACC Thr	AAC Asn	AGC Ser
AUA Ile	ACA Thr	AAA Lys	AGA Arg
AUG Met	ACG Thr	AAG Lys	AGG Arg
GUU Val	GCU Ala	GAU Asp	GGU Gly
GUC Val	GCC Ala	GAC Asp	GGC Gly
GUA Val	CGA Ala	GAA Glu	GGA Gly
GUG Val	GCG Ala	GAG Glu	GGG Gly

Table 2

Amino acid names	
Ala = alanine	Leu = leucine
Arg = arginine	Lys = lysine
Asn = asparagine	Met = methionine
Asp = aspartic acid	Phe = phenylalanine
Cys = cysteine	Pro = proline
Gln = glutamine	Ser = serine
Glu = glutamic acid	Thr = threonine
Gly = glycine	Trp = tryptophan
His = histidine	Tyr = tyrosine
Ile = isoleucine	Val = valine

- 2** Draw the complementary strand on the diagram below, then label each part of the nucleotide chain. The first nucleotide has been drawn for you.



- 3** List three differences between DNA and RNA.

- 4** What is a *nucleotide*? _____

- 5** What do you think would happen if m-RNA copied the DNA code incorrectly?



6 Use Figure 2 to work out all the different codes for each amino acid. The first row has been done for you.

Amino acid	Complete name	Possible codes
Val	Valine	GUU, GUA, GUG, GUC
Trp	Tryptophan	
Thr	Threonine	
Ser		
Pro		
Phe		
Met		
Leu		
Ile		
Glu		
Cys		
Asp		

7 The following amino acids could produce proteins for eye colour.

Lys-Val-Tyr-Ser-Pro-Leu-Ile → blue eyes

Lys-Val-Tyr-Pro-Ser-Leu-Ile → hazel eyes

Lys-Pro-Tyr-Pro-Phe-Ser-Ile → light brown eyes

Lys-Pro-Tyr-Val-Phe-Ser-Ile → dark brown eyes

Use Table 1 to help you decode the following DNA or RNA sequences to reveal the eye colour of each person. *Hint:* Their DNA sequence is their eye colour gene.

	Person 1	Person 2
DNA code	TTC GGG ATG CAC AAA AGG TAG	TTC CAT ATG GGC AGA AAC TAA
m-RNA complementary code	_____	_____
Protein		
Eye colour		

	Person 3	Person 4
DNA code	_____	_____
m-RNA complementary code	AAG CCU UAC CCG UUU UCC AUC	AAG GUA UAC UCU CCG UUG AUU
Protein		
Eye colour		

8 Sometimes changes happen to DNA. For example, the DNA code of Person 2 – Suzan – TTC CAT ATG GGC AGA AAC TAA has changed to the following code due to a flip in the DNA bases: TTC CAT ATG AGA GGC AAC TAA. This is called a mutation.

a Find the complementary m-RNA code of Suzan's new DNA code. _____

b Work out the protein chain. _____

c What colour are Suzan's eyes now? _____



Every cell in your body has DNA. The DNA is tightly packed and coiled in the nucleus of each cell. A section of DNA is called a gene. This gene is made up of a sequence of bases that code for a protein. Your body continually makes protein for growth and repair. During **protein synthesis**, only part of the DNA is read – the rest is 'switched off'.

The following activity will help you understand the complexity of protein synthesis. The process of protein synthesis has been written in a procedure format. Carefully read the procedure and complete the activity.

The process of protein synthesis

Aim: To simulate how protein is made.

Materials needed:

- strand of DNA, about one gene
- one messenger RNA molecule
- one ribosome
- many transfer RNA molecules

Method:

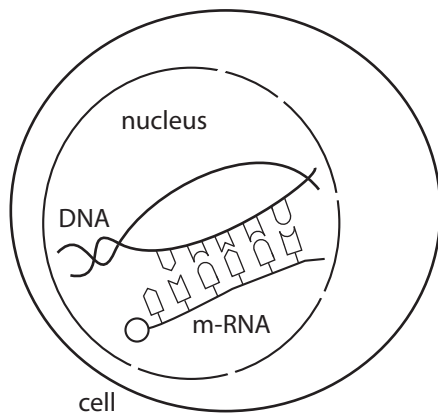
- Study the information below and the diagrams on the next page, which show the steps involved in protein synthesis in the human body.
- Cut out the steps and the diagrams.
- Arrange the diagrams in order from 1 to 8. Then match each information box to the correct diagram.
- Check your work with a classmate. When you agree on the order, paste the steps and diagrams in your workbook under the heading 'Protein synthesis'.

Extension

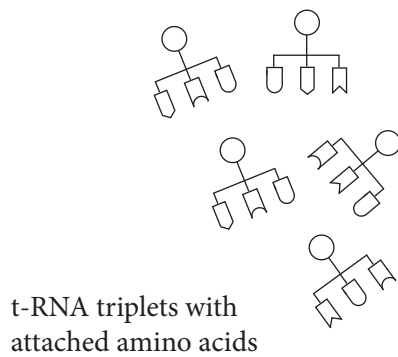
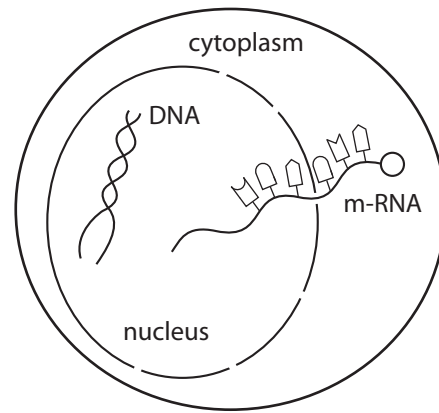
In your workbook, rewrite this procedure as a scientific explanation for how protein is synthesised in human cells. Use the procedure you have just ordered as a guide.



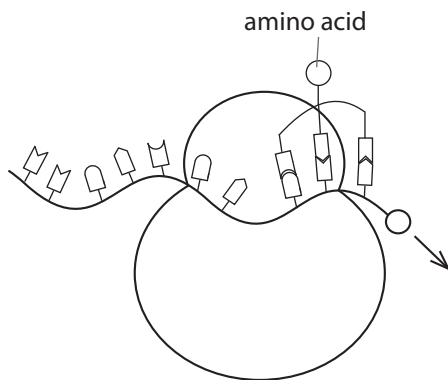
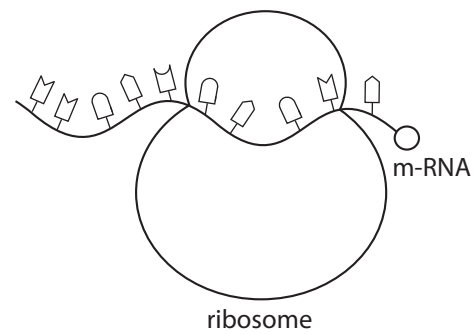
a Collect the transfer RNA molecules needed.	b Move the messenger RNA out of the nucleus into the cytoplasm.
c Add the second t-RNA onto the m-RNA and join each new amino acid to the previous one. Remove the first t-RNA as soon as the amino acid is connected to the second amino acid.	d Unzip the piece of DNA. Copy the code from one strand onto the messenger RNA by matching the bases adenine with thymine, and guanine with cytosine. (<i>Hint: RNA uses uracil instead of thymine.</i>)
e Connect the m-RNA onto a ribosome where the proteins will be assembled.	f Connect the t-RNA one at a time onto the m-RNA, matching to the triplet codes – three bases m-RNA.
g The completed protein should fall off.	h Repeat steps 5 and 6 for 200 amino acids until the stop code is reached on the m-RNA.



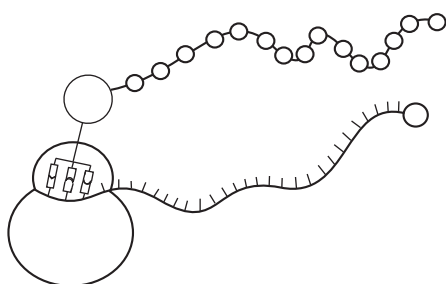
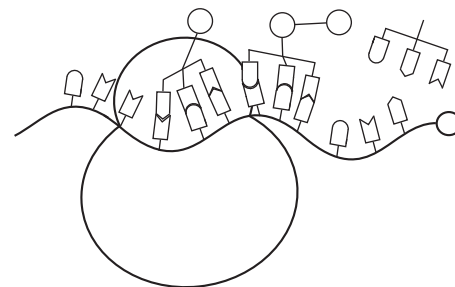
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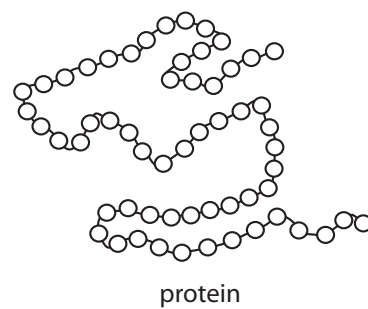
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7 8



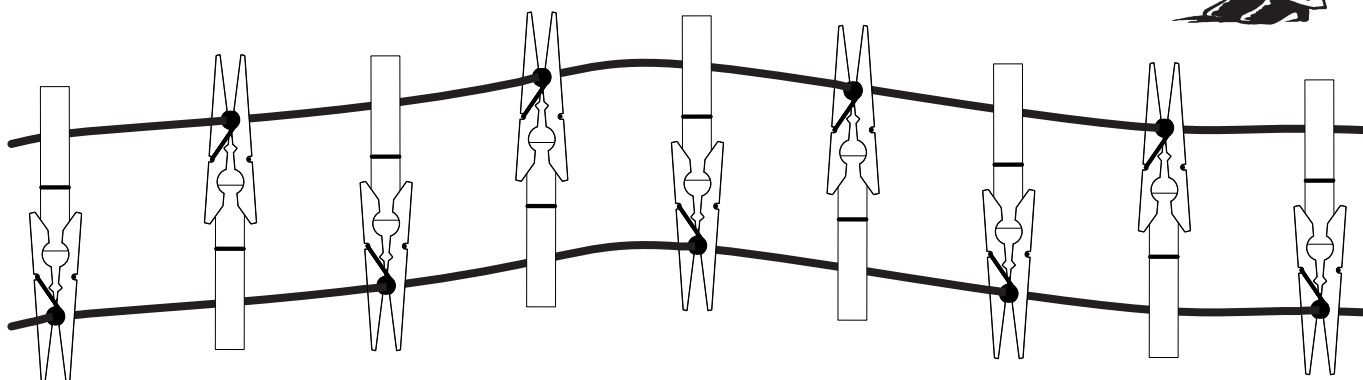


Recount

Yesterday Mrs De-enai asked us to investigate protein synthesis in cells. To do this, we constructed a model to show us how proteins are made. Each group was given a set of plastic pegs of five different colours – blue, green, yellow, red and white, wool in different colours, markers and paper.

Step 1

First we had to model transcription in the nucleus. To do this, we constructed a double strand of DNA. Our group decided that the red pegs would be adenine, blue pegs for thymine, yellow for guanine and green for cytosine. We threaded black coloured wool through the spaces in the pegs using the sequence CTTAAGATG, corresponding to the colours we had chosen. We then made a complementary strand of the DNA and pegged the two strands together. Refer to the diagram below.

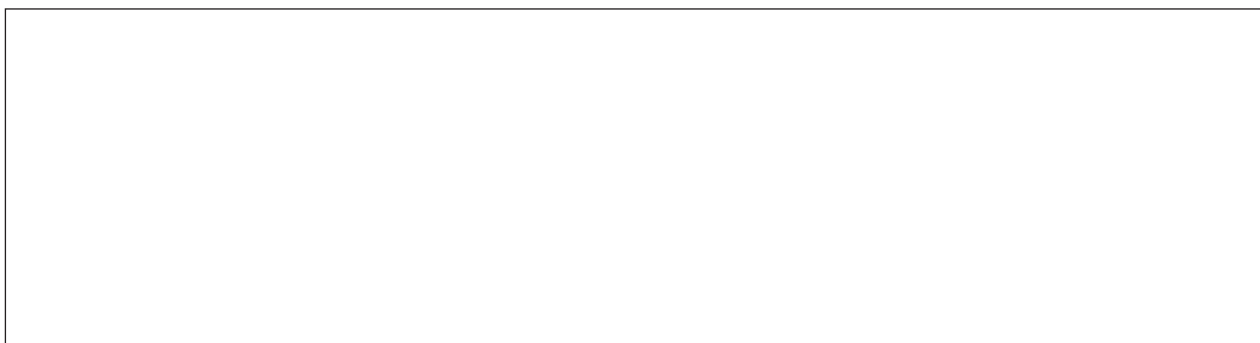


- 1 Write the colours of the pegs in the original DNA strand. Make sure the order of the colours is correct. Then colour the pegs to show this order.

Step 2

The next step was to unzip the double-stranded DNA molecule, producing two single-strand molecules. Using our original strand, we made a complementary molecule but this time we used red wool instead of black wool, and instead of thymine, our blue peg, we used uracil, a white peg. So we joined uracil, not thymine, to adenine. We called this complementary strand messenger RNA (m-RNA).

- 2 Draw, colour and label a diagram to show the m-RNA molecule.



Step 3

In this part of our investigation, we modelled translation. Translation occurs in the cytoplasm of a cell. The m-RNA unzips from the DNA and leaves the nucleus and attaches itself on a ribosome. Using the third coloured wool, green, we made three transfer RNA (t-RNA) molecules: CUU AAG AUG

We attached a small piece of cardboard, labelled with the amino acids' names in the correct order: Leucine, Lysine, Methionine. The three t-RNA molecules were attached to the corresponding m-RNA molecule.



- 3** Draw, colour and label a diagram to show the three t-RNA molecules attached to the m-RNA molecule.

We stuck the amino acids together with tape and removed the t-RNA pegs. The end result was a polypeptide chain made up of all the amino acids.

- 4** Draw, colour and label a diagram to show the polypeptide chain made up of all the amino acids.

- 5** Write down the original DNA sequence. _____
- 6** What was the complementary m-RNA sequence? _____
- 7** What were the t-RNA codes? _____
- 8** Describe in your own words what you have learned from this model. _____

- 9** Where does transcription take place? _____
- 10** Explain what happens to DNA during transcription. _____

- 11** Where does translation take place? _____
- 12** Explain what happens during translation. _____

- 13** The code on a section of DNA reads ATCGGCATATCTACT.
- a** What would the complementary m-RNA read? That is, what are the m-RNA codons? _____
- b** What would the anti-codons on t-RNA read? _____
- 14** Rewrite the recount as a procedural text type. Start with a heading, then include an aim, equipment list, method, results and a conclusion.
- 15** Use different coloured pegs and wool to model protein synthesis for yourself.
- a** Does this model help you understand protein synthesis? Explain.
- b** What are the limitations of this model?
- c** Suggest how you could improve this model of protein synthesis.



Read the model discussion below and then complete the tasks on the next page.

The Tasmanian tiger, *thylacine*, is a marsupial carnivore that used to live in Tasmania. European settlers believed this animal to be dangerous to them and their farm animals and so started hunting it down. The last thylacine died in captivity in 1936. Some scientists want to clone a thylacine and reintroduce it to Tasmania, but there are arguments against as well as in favour of this issue.

Thylacines were native animals of Tasmania. It was only because European settlers feared them that they were exterminated. Had the European people not murdered this indigenous marsupial, the thylacine would still be living in Tasmania today. Reintroducing cloned Tasmanian tigers would only make right the wrong done by people of the past.

Cloning technology for animals exists. Scientists know how to clone animals and have cloned many already: sheep, cows, goats, cats and mice are some examples. Although an extinct animal has not been cloned, with some more research scientists could do that too. There is no scientific technology standing in the way of cloning the Tasmanian tiger.

On the other hand, the Tasmanian tiger lived in Tasmania more than one hundred years ago. Since then, the large island has changed – there is less bushland, and more farms and towns. It is no longer the same Tasmania. Sending an extinct animal into today's Tasmania would be as thoughtless as expecting a polar bear to survive in the jungle.

When scientists clone animals, many cells with a new nucleus never develop into offspring, or the offspring may be malformed. There is a tremendous amount of tragic wastage during cloning. It is also very expensive. Even if a stem cell is filled with a thylacine nucleus, there are many questions that still need to be answered: Which animal would the stem cell come from? Which animal would carry the new cells and give birth to the infant Tasmanian tiger? Which animal would nourish the growing offspring? Until these issues have been resolved satisfactorily, cloning of the extinct thylacine should not be attempted.

In conclusion, the Tasmanian tiger should not be cloned. It is too expensive and wasteful and a positive outcome cannot be guaranteed. It is also likely that a reintroduced thylacine would die out again, because too much of Tasmania has changed significantly during the last century. Unfortunately, the Tasmanian tiger has become extinct and it should remain so.



- 1** The discussion text type on the previous page contains four sections: *Introduction*, *Arguments for*, *Arguments against*, *Conclusion*. Draw a large square bracket in the left margin to indicate each section, and label it.
 - 2** Often a paragraph includes key words near the start to describe what it is about. Double underline the key words in each paragraph.
 - 3** In the right margin, briefly summarise the main content of each paragraph.
 - 4** State the purpose of this discussion.
-
- 5** The arguments in a discussion are usually backed up with facts. Scientific words can highlight these. Circle at least five scientific words in the arguments section.
 - 6** Although discussions contain many facts, they also use emotive language that exaggerates and appeals to your feelings. One example of emotive language is ‘murdered’. Underline at least five other examples.
 - 7** Select one topic from the list below and write a discussion about it. Start by planning useful points for each paragraph. You may need to research the topic first. In the scaffold at the bottom of the page, note useful points you could include for each paragraph. Then in your workbook write your discussion in six paragraphs.

Topics

- a** In the USA, people can pay \$50 000 to have their favourite dead cat cloned. Is that wise?
- b** A farmer wants her ageing prize ram cloned so that she can continue to use it for breeding.
- c** Phar Lap was a very fast Australian racehorse and he died in 1932. He should be cloned so he can win horse races again.
- d** A cow producing unusually large amounts of milk is going to be cloned 12 times to replace a large but old cow herd.
- e** More mice should be cloned so that scientists can learn more about cloning techniques.

Discussion scaffold

1. Introduction	2. First argument for	3. Second argument for
4. First argument against	5. Second argument against	6. Conclusion and recommendation



Read the text about cloning and then complete the task on the next page.

Explanation of cloning

A clone is an organism which is genetically identical to another organism. Cloning is a form of asexual reproduction which is widespread in nature. For unicellular organisms and many plants, it is a normal process, such as cell division or vegetative propagation. Mammals can produce identical twins when very young embryos split in two.

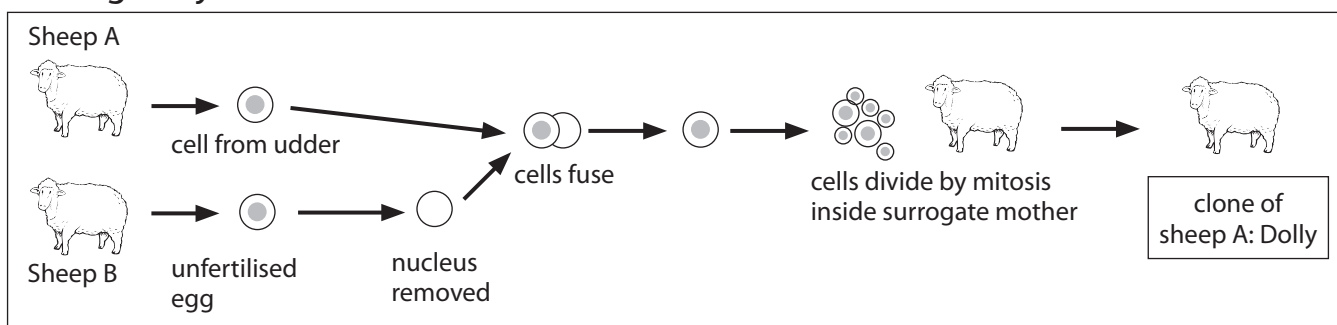
However, artificial cloning is the cause of much debate. There are two types of artificial cloning: embryo splitting and nuclear transfer.

- **Embryo splitting** is similar to the production of twins or multiple births, except it does not happen naturally – it is brought about by a scientist or doctor. In this type of cloning, the clones are the same age.
- During **nuclear transfer**, the complete set of chromosomes of one individual (the genetic parent) is transferred into a stem cell of another individual (the cell donor parent) and carried by a third individual (the host parent). This method allows the chromosomes to be transferred at different times and so the clones can be of completely different ages. For example, a pet cat that has died could be cloned for its owners.

It is also possible to transfer only part of a chromosome of an individual to a complete set of chromosomes of another individual. For example, a strawberry could carry a fish gene to prevent it from freezing at very low temperatures. This method produces **transgenic species**, which could not be produced in nature.

While embryo splitting has a fairly high success rate, nuclear transfer has often resulted in the death of the new clones or in new clones with health or development problems. Dolly, the cloned sheep, was one survivor of 300 clones. The diagram below outlines the process involved in producing Dolly the cloned sheep.

Cloning Dolly



Applications of cloning

In the process of gene farming, transgenic bacteria or animals are used to produce medicines. The advantages of medicines produced by biogenetic manufacturing processes are that they can be obtained in a much purer form than with conventional techniques that involve animal and human intermediate products. Production can be on a large scale and relatively cheap. However, there are also risks to the bacteria and animals due to their transgenic manipulation. People taking those medicines may not respond favourably and could become sick. Careful testing of drugs produced by gene farming is necessary.

In gene targeting or gene knock-out, individuals can be given favourable genes or have unfavourable genes destroyed. This would result in healthier offspring or offspring with certain characteristics. For example, cows could produce milk with a higher protein content. Using this process on animals would also help scientists learn how to cure genetic diseases in people.

During tissue cloning, particular human tissues are grown from the patient's cells rather than using donated tissue or animal tissue. The theory is that the patient's immune system would not reject the cloned tissue because it would not be made from foreign cells. For example, the burnt skin of a patient could be replaced with the patient's own but cloned skin.

Another idea for cloning human tissues involves swapping animal genes with human genes so that the animal grows a particular human organ. This organ would then be transplanted into the human.



Artificial insemination of farm animals, during which a female animal is fertilised with collected sperm from a particular male, has been occurring for a long time. However, it is a slow process. Usually only one valuable offspring is born at a time. If the chromosomes of a desirable animal could be used to grow clones, many more valuable offspring could be produced. This could result in great commercial and economic advantages. Even dead animals could be cloned.

Concerns about cloning

When cloning occurs in nature, more of already existing organisms are produced. This also happens during embryo splitting and complete nuclear transfer. However, artificial cloning is expensive and wasteful, although it could produce desirable and valuable organisms. As scientists practise cloning, they learn more about genetics, cloning techniques and development. Yet somebody must decide what is desirable and valuable, who should own those organisms, and what happens to the 'old' organisms.

Transgenic species are organisms that do not exist in nature; they are artificially made. It is not known what effect transgenic species could have on nature, if they were released. Once they are released, it may not be possible to capture them again. Potentially, they could interbreed and cause disease and destruction. Many questions about the possible consequences of cloning remain unanswered.

Task

Select one type of cloning and brainstorm possible applications and examples. You could use a mind-map to help organise your ideas. Then write a discussion:

- Start with an introduction in which you describe and explain your type of cloning.
- Develop two elaborated arguments which explain why that type of cloning should occur, and also develop two elaborated arguments against that type of cloning occurring.
- Finally, make a recommendation based on your arguments.

Don't forget to use persuasive language such as *should, must, always, never, only, rarely, extremely, under no circumstances* in order to convince the reader. Use conjunctions such as *on the other hand, however, as well as, but, since, when, because* to link your thoughts together.

To help you structure your discussion, use the scaffold below. Note useful points you could include for each paragraph. These notes will form the basis for your written discussion. Then in your workbook, write your discussion in six paragraphs.

Discussion scaffold

1. Introduction	2. First argument for	3. Second argument for
4. First argument against	5. Second argument against	6. Conclusion and recommendation



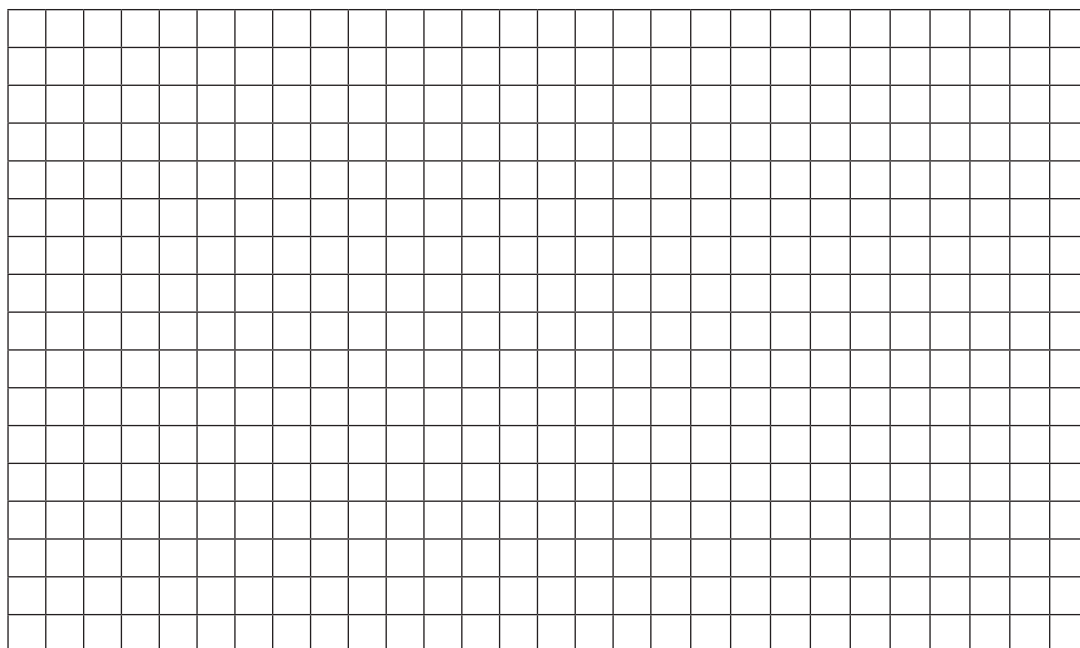
Has anyone ever told you that you have your father's nose or your grandmother's hair? Many of our features are inherited from our parents, and theirs from their parents. **Inheritance** or **heredity** is the passing of **characteristics** from parent to offspring – you. Characteristics include eye colour, hair colour, height, nose shape – and many more. We all inherit slightly different combinations of these features from our parents. Even brothers and sisters look different, because they inherit different characteristics from their parents.

- 1** The table below shows data for characteristics collected from 30 year 9 students. Some of the 30 students did not respond for some characteristics. Add up each tally and calculate the percentage of students with that variation in characteristic. One example has been started for you. *Hint:* The percentages for each variation of a particular characteristic should add up to 100; e.g. $73\% + 27\% = 100\%$.

Characteristic	Variations in characteristic	Tally	Tally total	Total number of students	Percentage
Tongue rolling	Yes, can		22	30	$(22 \div 30) \times 100 = 73\%$
	No, cannot			30	$(8 \div 30) \times 100 = 27\%$
Attached earlobes	Yes, attached				
	No, unattached				
Receding hair on the sides of the forehead	Yes, present				
	No, not present				
Gap between front teeth	Yes, gap				
	No, no gap				
Right thumb on top during hand folding	Yes, right thumb on top				
	No, left thumb on top				
Hair on the second digits of your fingers	Yes, present				
	No, not present				
Right dominant hand	Yes, right				
	No, left				
Second toe longer than big toe	Yes, longer				
	No, shorter				
Straight hair	Yes, straight				
	No, not straight				



- 2** Use the percentages you calculated on the previous page to construct a column graph of the results on the grid below.



- 3** Make two general statements about the graph.

- 4** Which of the more frequent variations do you possess?

Sometimes characteristics can be influenced by the environment. For example, consider two people with almost identical skin colour – one person likes to be outdoors and the other prefers to spend time indoors. These environments will affect their skin colour. Eating habits can impact on how tall people grow. Poor nutrition can inhibit growth, therefore a person with poor eating habits might not reach full height. Even identical twins can appear different if they are raised in different environments.

- 5** Which characteristics in the table on the previous page can be altered or influenced by the environment? Explain how.

- 6** Think about your own development. Do you think it has been influenced more by the environment or more by inheritance? Justify your answer using examples.



All organisms are made of cells, and each type of organism has a particular number of chromosomes. For example, each body cell of a dog has 28 chromosomes, while each body cell of a horse has 64 chromosomes. Human cells contain 46 chromosomes, or 23 pairs of chromosomes, or 2 sets of 23 chromosomes ($= 2n$).

Reproductive cells – sperm in males and eggs (ova) in females – always contain half the number of chromosomes of the body cells; that is, one set of 23 chromosomes ($= 1n$). This ensures that the baby also has 46 chromosomes, not 92. When an egg and sperm join during fertilisation, 23 chromosomes come from the mother and 23 come from the father. The fertilised egg has a complete set of 46 chromosomes; that is, 23 pairs of chromosomes.

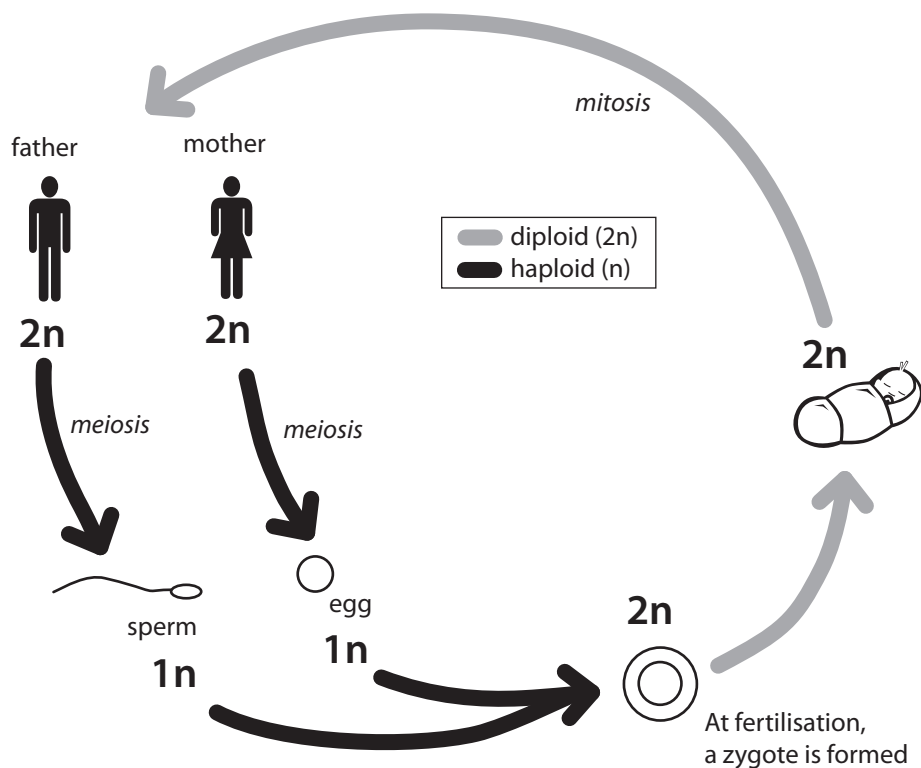


Figure 1

1 What do you think would happen if the egg and sperm always had a complete set of chromosomes?

2 Find the terms 'diploid' and 'haploid' in figure 1. What do you think these terms mean?

A special pair of chromosomes determines whether the fertilised egg will be a boy or a girl. This pair is made up of the sex chromosomes. In males, one chromosome in this pair is smaller. The larger chromosome is called X and the smaller is called Y. Females have two X chromosomes, which are the same size. Males have X and Y. These X and Y chromosomes are called sex chromosomes, because they contain the genes that determine sex. The other chromosomes are called autosomes, and these include all the other pairs of chromosomes. Each pair of autosomes in the human karyotype is numbered.

Figure 2 shows the karyotype of a human female and a human male. The first 22 pairs of chromosomes are the autosomes. The 23rd pairs are the sex chromosomes.

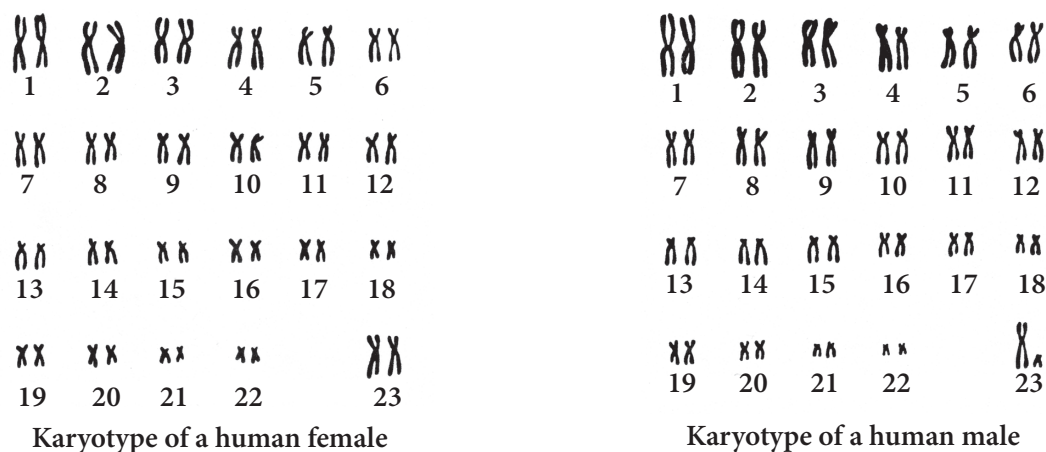


Figure 2



As shown in figure 3, when a sperm with an X chromosome joins with an egg, the child will be a girl (XX). When a sperm with a Y chromosome joins with an egg, the child will be a boy (XY).

Males produce equal numbers of X and Y sperm, which means the chance of producing a boy or a girl is 50:50.

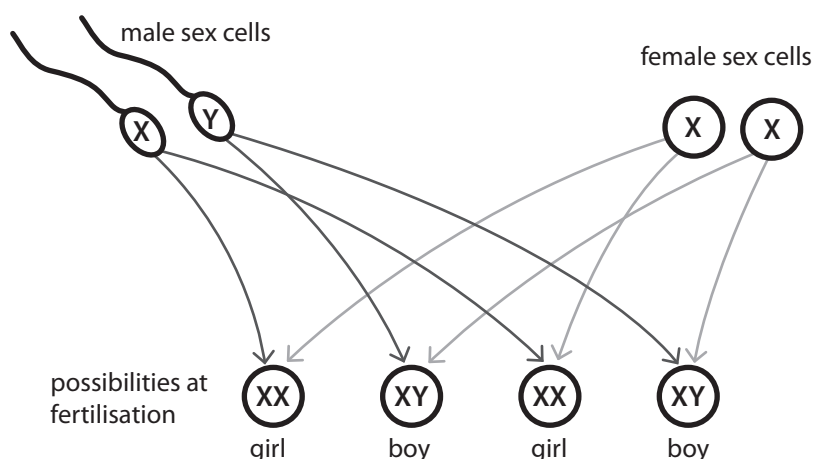


Figure 3

3 Define these terms.

a Autosome

b Karyotype

4 How many chromosomes do humans have in:

a skin cells?

b liver cells?

c sperm?

d ovum?

e nerve cells?

5 Describe how the sex of a human baby is determined.

6 Rewrite the summary points in the table in sentence form.

Location	Cell division	Description	Purpose
Somatic (body) cells	Mitosis	$2n$ (diploid) $\rightarrow$ $2n$ (diploid)	Growth and repair
Sex organs	Meiosis	$2n$ (diploid) $\rightarrow$ n (haploid)	Gamete production

7 A karyotype of a human baby is shown at the right.

a What sex is the baby? Explain how you know.

b How many chromosomes does the baby have?

c How many chromosomes did the mother pass on to baby?

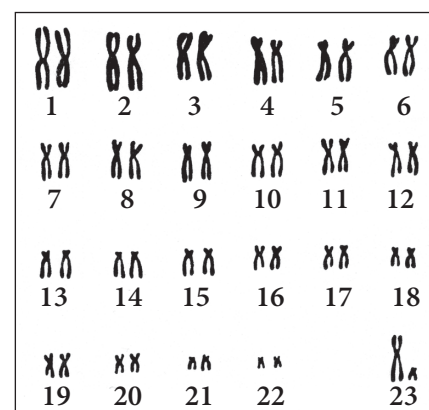
d How many chromosomes did the father pass on to baby?

e How many chromosomes are there in the mother's skin cells?

f How many chromosomes are there in the mother's eggs?

g How many chromosomes are there in the father's sperm?

h How many chromosomes are there in the father's skin cells?



Karyotype of a human baby

8 a King Henry VIII reigned in England from 1509 to 1547 and had six wives. Historians believe that Henry beheaded his wives when they delivered baby girls, because he wanted a son to continue the royal succession. He blamed his wives for producing daughters instead of sons. Was Henry correct?

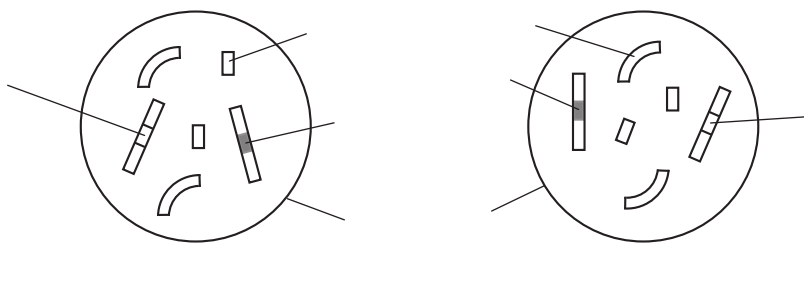
Who determines the sex of a baby? Explain your answer.

b Why do you think this information was not known at the time of Henry VIII?

**Rabbit genes**

Imagine a group of rabbits. Each one has a gene for dark fur and a gene for light fur.

- 1** Simplified diagrams of body cells from a female rabbit and a male rabbit are shown below. Use the word bank to label the diagrams.

**Word bank**

- chromosome
- light fur gene
- dark fur gene
- light fur gene
- dark fur gene
- cell membrane
- Female body cell
- Male body cell

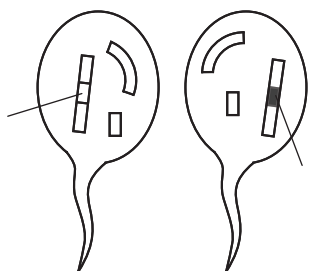
- 2** Refer to the diagrams of the body cells above to answer the following questions.

- How many chromosomes are in each body cell? _____
- Does each chromosome have a matching partner? _____
- How many pairs of chromosomes are there in each body cell? _____
- If all the rabbits have dark fur, which gene is dominant and which recessive?

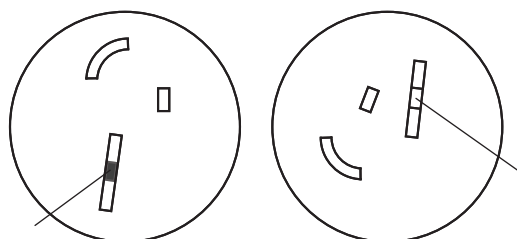
- 3** Rabbits actually have 44 chromosomes in each body cell, but only three pairs of chromosomes have been shown in the diagrams above for the sake of simplicity.

- During meiosis, gametes (sex cells) are produced. Male gametes are called _____ and female gametes are called _____.
- In the simplified version of rabbit body cells shown in the diagrams above, how many chromosomes would each gamete have? _____
- Is it possible for two male gametes to have different fur colour genes? _____
- Is it possible for two female gametes to have different fur colour genes? _____
- Out of 100 eggs, how many would have a dark fur gene? _____
- Out of 100 eggs, how many would have a light fur gene? _____

- 4** Simplified diagrams of sex cells from a female rabbit and a male rabbit are shown below. Use the word bank to label the diagrams.

**Word bank**

- light fur gene
- dark fur gene
- light fur gene
- dark fur gene
- cell membrane
- Types of sperm
- Types of eggs



- 5** Is there any way of predicting which type of sperm will fertilise each type of egg? _____

- 6** In your opinion, how many of 100 baby rabbits from the above parents would have:

- dark fur? _____
- light fur? _____



Most people know the saying ‘breed like rabbits’ – rabbits produce so many offspring! But what do the baby rabbits look like? Say two rabbits with dark fur and genes for dark as well as light fur ‘breed like rabbits’ and produce one hundred offspring. What would these babies look like? What would their genes for fur colour be?

This activity is easier to do and understand if you have completed topic 11, ‘Rabbits reproducing’.

To complete this activity you need scissors, two pieces of card or paper measuring 10 cm × 10 cm, one light coloured and one dark coloured, and two bags or other small containers.

Work in small groups of 2 to 4 students for this activity.

Preparing light and dark genes

- On light coloured card, draw up a 1 cm grid measuring 10 cm × 10 cm. This grid represents 100 light fur genes.
- On dark coloured card, draw up the same kind of grid. This one represents 100 dark fur genes.
- Cut out the 100 dark squares and the 100 light squares and keep them separate from each other.
- Place 50 light squares and 50 dark squares into a bag and label the bag ‘sperm’. Place the other 50 light squares and 50 dark squares into another bag and label it ‘eggs’. Shake each bag to mix the gametes thoroughly.

Producing offspring

- Take one ‘egg’ at random (that is, without looking) out of the egg bag and place it on the table.
- Take one ‘sperm’ at random and place it on the table next to the ‘egg’.
- Repeat, until all 100 eggs and 100 sperm are paired.
- Sort the egg–sperm pairs into matching colour groups. Altogether, there should be 100 offspring from the two parents.

1 Count the number of pairs in each colour group and record your results in the Numbers row of the table.

Aspect	Dark–dark genes	Dark–light genes	Light–light genes
Numbers			
Divide by smallest number	÷	÷	÷
Ratio	:		

2 To calculate the ratio of gene combinations, in the next row of the table, divide the total number for each colour group by the smallest of the total numbers and round your answer to two decimal places. Then write the answers in the ratio row.

3 Collect the numbers of pairs in each colour combination from the other groups in your class. List them in the first row of the table, and add them to give the total numbers for the class. Then calculate the ratio of the whole class numbers.

Aspect	Dark–dark genes	Dark–light genes	Light–light genes
Results of other groups			
Total numbers for class			
Divide by smallest number	÷	÷	÷
Ratio	:		

4 Which ratio is closer to whole numbers – your group ratio or the class ratio? _____

5 Based on your class results, if there are four baby rabbits, how many rabbits would have:

a a dark–dark gene pair? _____ **b** a dark–light gene pair? _____ **c** a light–light gene pair? _____



6 Based on your class results, if there are four offspring, how many would have:

a dark fur? _____

b light fur? _____

c grey fur? _____

Hint: Remember that dark is dominant over light for rabbit fur.

Theoretical prediction

7 a Dominant genes are shown with a capital letter, usually the first letter of the particular gene (i.e. D). Recessive genes use the same letter but they have a small letter (i.e. d). Write the gene letters for the characteristics in the Punnett square at the right.

b Now 'multiply' the egg genes with the sperm genes to find the offspring gene combinations. Write the combinations in the Punnett square.
Hint: The dominant gene is written first, then the recessive gene.

		Eggs	
		Dark =	Light =
Sperm	Dark =		
	Light =		

8 What is the offspring ratio of dark-dark (DD) : dark-light (Dd): light-light (dd) gene pairs?

_____ : _____ : _____

9 What is the offspring ratio of dark offspring : light offspring? _____ :

Evaluation and application

10 How could the experimental results of the ratio be brought closer to the theoretical prediction?

Every time a rabbit is born, there is a 3 in 4 chance (i.e. 75%) that it has dark fur and a 1 in 4 chance (i.e. 25%) that it has light fur.

11 If two rabbit parents have one offspring, what colour is it most likely to be? _____

12 If two rabbit parents have three dark offspring already, what is the most likely colour of:

a the fourth offspring? _____

b the tenth offspring? _____

c the fiftieth offspring? _____

13 If two rabbits have 200 offspring in their life time, according to the theoretical prediction:

a how many would have dark fur? _____

b how many would have light fur? _____

c how many would be pure-bred dark – that is, have a dark-dark (DD) gene pair? _____

d how many would be hybrid dark – that is, have a dark-light (Dd) gene pair? _____

14 Is a purebred dark rabbit darker than a hybrid dark rabbit? _____

15 Is it possible to have a purebred and a hybrid light fur rabbit? _____

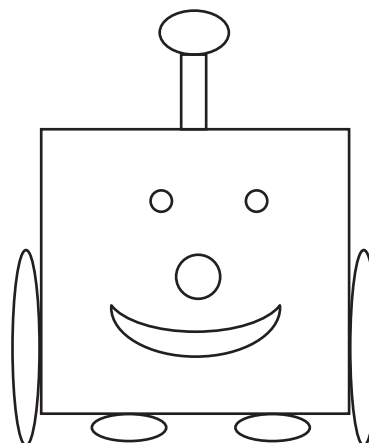
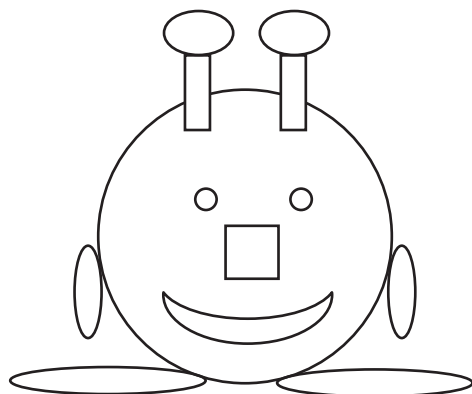
16 Is a rabbit with light fur always purebred? _____

17 Is a rabbit with dark fur always purebred? _____



The Beeple people from the planet Genetoz have volunteered to show you their chromosomes to help you develop a clearer understanding of how inheritance works.

Pictured below are Mum and Dad Beeple. Mum is expecting a baby. By using Mum and Dad's characteristics, you will be able to predict what their baby may look like.



Describing Mum's and Dad's looks: Phenotype

Mum Beeple has the most amazing brown eyes, a cute orange nose and elegant long blue feet. What attracted her to Dad was his bright green eyes. He too has an orange nose, but his best feature is his short red feet.

- 1** Colour Mum and Dad to show what they look like – that is, their phenotypes.

Describing Mum's and Dad's genes: Genotype

Look at the characteristics decoder table at the right. Remember, every characteristic is determined by two genes. The dominant gene, shown by a capital letter, usually 'covers up' the recessive gene, which is shown by a small letter.

- 2** Use the characteristics decoder table to find out which gene pair (genotype) matches each appearance (phenotype). Complete the genotype table below, which shows the genes Mum and Dad have for each trait. Some genes have been included for you.

Characteristics decoder table

Trait	Appearance (phenotype)	Gene pair (genotype)
Sex	Male: square Female: round	XY XX
Eye colour	Brown Green	BB or Bb bb
Nose shape	Square Round	SS or Ss ss
Nose colour	Red Orange Yellow	RR Rr rr
Feet length	Long Short	LL or Ll ll
Feet colour	Purple Red Blue	PP Pp pp
Arm length	Long Short	AA or Aa aa
Number of antennae	One antenna Two antennae Three antennae	NN Nn nn

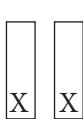
	Trait							
	Sex	Eye colour	Nose colour	Nose shape	Feet length	Feet colour	Arm length	Number of antennae
Mum		BB		Ss	Ll			
Dad							Aa	



Describing Mum's and Dad's chromosomes: Karyotyping

3 The chromosomes for Mum and Dad have been sorted into matching pairs below. Each gene for a characteristic occurs on a different chromosome. Write one letter onto each chromosome to show which gene is present. Some genes have been included for you.

Mum's genes:



Sex



Eye colour



Nose colour



Nose shape



Feet length



Feet colour



Arm length

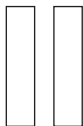


Antenna number

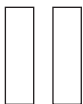
Dad's genes:



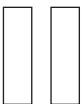
Sex



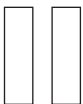
Eye colour



Nose colour



Nose shape



Feet length



Feet colour



Arm length



Antenna number

Making and describing Baby

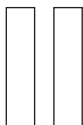
4 It's time to predict what Mum and Dad's baby may look like. Follow the steps below.

- Work in pairs, with one person representing Mum and the other person representing Dad.
- Look at the first chromosome pair from Mum's and Dad's gene charts above. Each person should flip a coin to choose one gene at random from their pair. Record the letter of the gene in Baby's chart below.
- Repeat this procedure, choosing chromosomes at random from the other seven chromosome pairs.

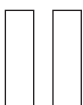
Baby's genes:



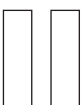
Sex



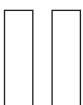
Eye colour



Nose colour



Nose shape



Feet length



Feet colour



Arm length



Antenna number

5 In the table below, show Baby's genotype by referring to the chromosome pairs above. Then refer to the characteristics decoder table on the previous page to identify Baby's phenotype.

Baby's traits	Sex	Eye colour	Nose colour	Nose shape	Feet length	Feet colour	Arm length	Number of antennae
Genotype								
Phenotype								

6 Use the illustrations of Mum and Dad Beeple on the previous page to draw and colour a picture of Baby.

7 How many chromosomes:

a do Mum and Dad have? _____ **b** does Baby have? _____

8 Does your baby look like Mum or Dad? _____

9 Compare babies with the students next to you. Are they similar? _____

10 Does anyone in your class have a baby:

a identical to yours? _____ **b** exactly the opposite of yours? _____

11 Challenge! Work in small groups to figure out how many possible gene combinations there are for Baby.



Pedigrees or family trees are useful tools for showing patterns of **inheritance** across a family. Pedigrees use symbols and shadings to communicate information. Some of the symbols used are shown below.



male



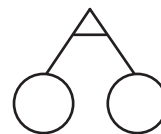
female



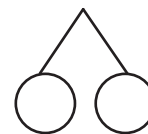
male, deceased



female, deceased



identical twins



fraternal twins

relationship/marriage
line

descent line



sibling line

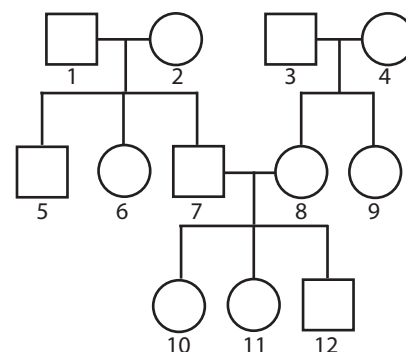
male with feature
being observedfemale with feature
being observed

1 Look at the pedigree at the right.

- How many females are there? _____
- How many males are there? _____
- How many grandchildren are there? _____
- Are there any twins in the family? _____
- What is the relationship between individuals 7 and 8? _____
- What is the relationship between individuals 5 and 10?

- What is the relationship of individuals 3 and 4 to individual 11?

- All females have the feature of brown eyes. Show that in the pedigree.



2 Use the starter pedigree below to draw your own family tree.

Tips for what to include at each level are listed at the left. Make sure you use the correct symbols.

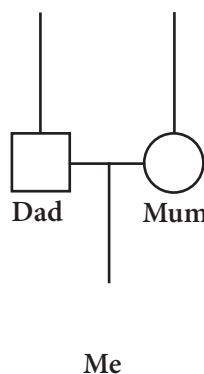
Great-grandparents

Grandparents

*Parents, aunts,
uncles*

*Sisters, brothers,
cousins*

Nieces, nephews



3 Choose a feature that runs in the family, like tongue rolling or a widow's peak. Shade the people in your family tree that have this feature.



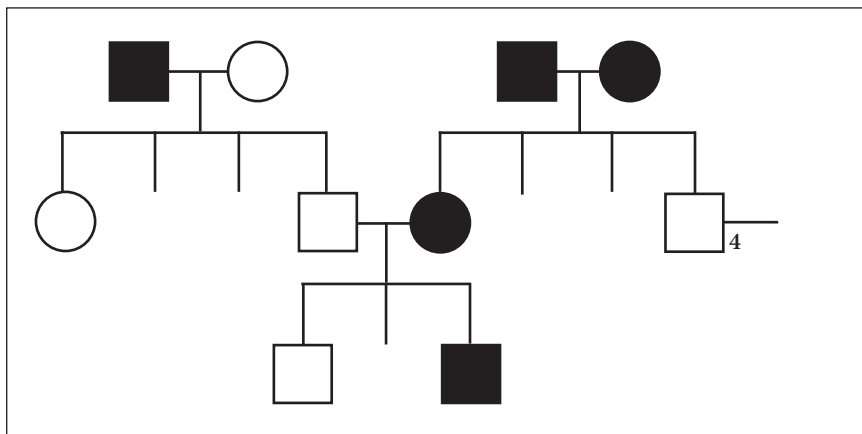
You need a partner for this activity. One person should have Sheet A, and the other person should have Sheet B. Do not show your sheet to your partner!

Your pedigrees and those of your partner are incomplete. Your challenge is to complete the pedigrees, **without** seeing your partner's diagrams. To do this, you will need to ask your partner careful questions about the number of people in each generation, their sex, their relationships to each other and whether they have the feature being observed or not. You need to complete your pedigrees before you can answer the questions.

Have fun, and remember – do not let your partner see your pedigree diagrams!

Tongue rolling

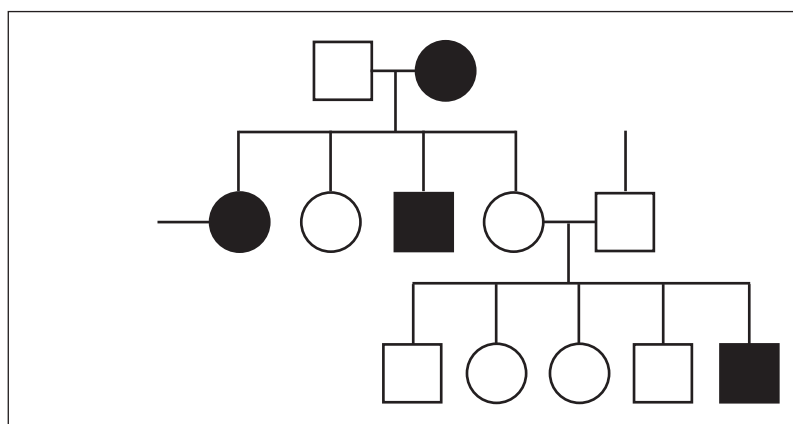
The pedigree shows three generations of a family. The shaded shapes represent individuals who can roll their tongue. Remember, work with your partner to complete the pedigree before you answer the questions.



- 1** In this pedigree:
 - a** how many females are there?
 - b** how many males are there?
 - c** how many grandchildren are there?
 - d** how many females can roll their tongue?
 - e** how many males in the second generation can roll their tongue?
- 2** Why does couple 4–5 have a child who cannot roll their tongue?
- 3** Is tongue rolling a dominant feature? How can you tell?

Polydactyly

The pedigree shows *polydactyly* which is an inherited characteristic. A person with this feature is born with an extra finger or toe.



- 4** In this pedigree:
 - a** how many females are not affected?
 - b** how many females are affected?
 - c** how many grandchildren are shown?
 - d** how many grandchildren have inherited the characteristic?
- 5** Is this feature dominant or recessive? Explain.
- 6** How many generations are shown in this family?
- 7** Now draw a family pedigree in which the children born are in the order girl, boy, boy. The youngest child is married and has two children of his own, both girls. The mother of these two girls has two younger brothers.
- 8** Mary has attached earlobes but her mother and father have free earlobes. Mary's grandmother (her father's mother) and Mary's grandfather (her mother's father) have attached earlobes. Draw Mary's family pedigree, showing attached earlobes. Is this a dominant or recessive characteristic? How do you know?



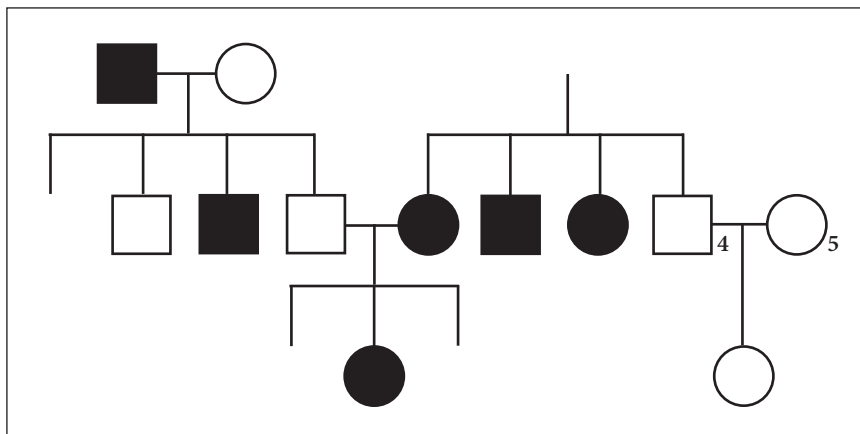
You need a partner for this activity. One person should have Sheet A, and the other person should have Sheet B. Do not show your sheet to your partner!

Your pedigrees and those of your partner are incomplete. Your challenge is to complete the pedigrees, **without** seeing your partner's diagrams. To do this, you will need to ask your partner careful questions about the number of people in each generation, their sex, their relationships to each other and whether they have the feature being observed or not. You need to complete your pedigrees before you can answer the questions.

Have fun, and remember – do not let your partner see your pedigree diagrams!

Tongue rolling

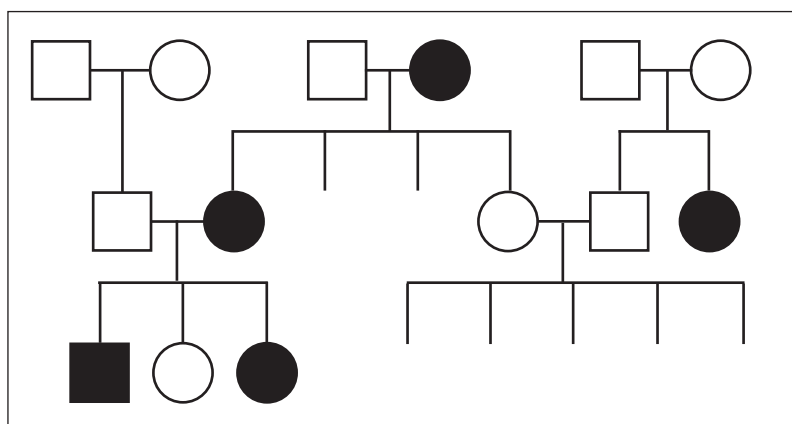
The pedigree shows three generations of a family. The shaded shapes represent individuals who can roll their tongue. Remember, work with your partner to complete the pedigree before you answer the questions.



- 1** In this pedigree:
 - a how many females are there?
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 - c how many grandchildren are shown?
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- 7** Now draw a family pedigree in which the children born are in the order girl, boy, boy. The youngest child is married and has two children of his own, both girls. The mother of these two girls has two younger brothers.
- 8** Mary has attached earlobes but her mother and father have free earlobes. Mary's grandmother (her father's mother) and Mary's grandfather (her mother's father) have attached earlobes. Draw Mary's family pedigree, showing attached earlobes. Is this a dominant or recessive characteristic? How do you know?



Many scientific terms have their origins in Greek or Latin words. The list below shows some terms used in genetics, their origins (Greek or Latin) and their meanings.

1 Draw an asterisk (*) in front of the terms in bold that are nouns, and underline the verbs.

- | | | |
|--|---|--|
| • allellon (Greek) <i>mutual, other</i> | • genesis (Latin) <i>birth, creation</i> | • parens (Latin) <i>mother or father</i> |
| • allos (Greek) <i>other</i> | • genos (Greek) <i>origin, race</i> | • phainein (Greek) <i>to show</i> |
| • chroma (Greek) <i>colour</i> | • helix (Greek) <i>spiral, twisted thing</i> | • plasma (Greek) <i>something moulded or formed</i> |
| • dominare (Latin) <i>to rule, to be master</i> | • heteros (Greek) <i>other</i> | • recessus (Latin) <i>retreat, a going back</i> |
| • filia (Latin) <i>daughter</i> | • homos (Greek) <i>same</i> | • soma (Greek) <i>body</i> |
| • filius (Latin) <i>son</i> | • hybrida (Latin) <i>mongrel</i> | • typos (Greek) <i>model, image, form</i> |
| • genea (Greek) <i>breed</i> | • kytos (Greek) <i>hollow vessel</i> | • typus (Latin) <i>image, form</i> |
| • generare (Latin) <i>to produce</i> | • morphe (Greek) <i>shape</i> | • zygotes (Greek) <i>yoked, joined</i> |
| • genesis (Greek) <i>origin, creation</i> | • nucleus (Latin) <i>kernel, nut</i> | |

2 Use the information above to help you write a definition for each of these scientific terms.

- | | | |
|---|--|---|
| a allele (short for allelomorph) | b chromosome | c cytoplasm |
| d dominant | e gene | f F generation or filia generation |
| g genotype | h helix | i heterozygous alleles |
| j homozygous alleles | k hybrid | l nucleotide |
| m nucleus | n P generation or parent generation | q zygote |
| o phenotype | p recessive | |

3 Use terms from question 2 to label the diagram of a cell.

4 Look at the chromosomes in the cell diagram and label the nucleus as heterozygous or homozygous.

5 Is a heterozygous cell about genotype or phenotype?

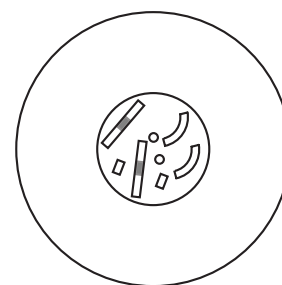
6 Identify two alleles from these genes: green eye colour, tall body size, brown hair colour, short body size.

7 An animal has a gene for black fur and a gene for white fur, and looks black. State:

- | | | |
|--|------------------------|--|
| a the phenotype. | b the genotype. | c the dominant gene. |
| d the recessive gene. | e the alleles. | f whether the animal is a hybrid. |
| g whether the animal is heterozygous or homozygous. | | |

8 Use terms from question 2 to complete the text.

The control centre of a cell is the _____, which is surrounded by _____. Inside it there are long, thread-like, dark-coloured structures called _____. Each chromosome is made up of the chemical DNA, which looks like a double _____. Each DNA molecule is made up of smaller units called _____. There are only four different nucleotides; the DNA molecules are different because they have different amounts of nucleotides and the nucleotides are in different sequences. Several _____ together represent a gene and all the _____ together make up a _____. The lengths and numbers of chromosomes vary for different organisms.



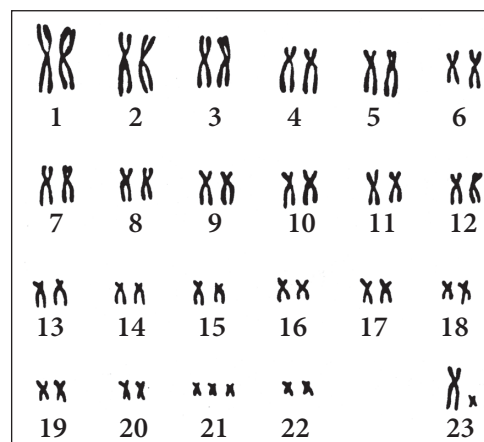


Genetic diseases occur when there is not the right number of chromosomes or when genes on chromosomes are not arranged correctly. Chromosomal disorders can occur when chromosomes fail to separate during the first meiotic division. As a result, one gamete may contain both chromosomes of one pair and the other has none. Deletions can also occur when a whole chromosome or part of a chromosome becomes lost. Radiation, some chemicals and excessive heat may cause mistakes to occur during meiosis.

The diseases described below are caused by chromosomal abnormalities.

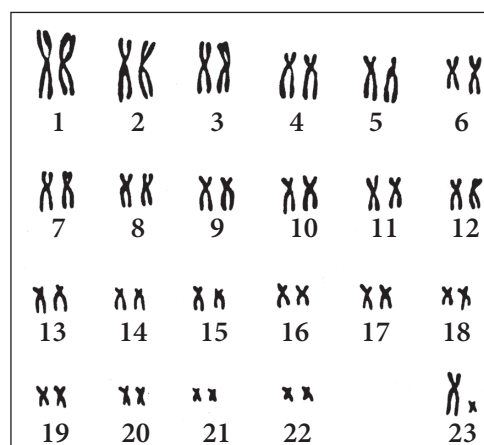
Down Syndrome

This disorder can be recognised by these characteristics: a short stature, stubby fingers, an eyelid fold, a wide gap between the first and second toes, a round head, a large tongue. The large tongue causes difficulties with speaking clearly. Another symptom is mental retardation. Down Syndrome is also called Trisomy 21, because there are three copies of the 21st chromosome, instead of two. The likelihood of a woman having a child with Down Syndrome increases rapidly with the age of the woman, from about 35 years. The frequency of Down Syndrome is about 1 in 800 births for women under 40 years of age and 1 in 80 for those over 40 years.



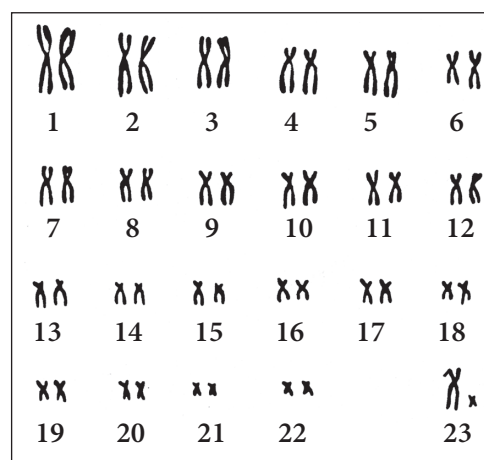
Cri du Chat Syndrome

A chromosome deletion is responsible for this disorder, which is also known as 'cat's cry syndrome'. Part of one of the 5th chromosomes is missing. An infant with this syndrome has a small head and a moon face. The baby's cry sounds like the meow of a cat, because of a malformed larynx. An older child has misshapen ears which are placed low on the head, and an eyelid fold. As the child matures, mental retardation becomes evident. Cri du Chat Syndrome occurs in about 1 of 50 000 births.



Fragile X Syndrome

With this condition, one X chromosome is nearly broken, leaving the tip hanging by a thread. Children with Fragile X syndrome appear to be normal, however they may be hyperactive or withdrawn. Their speech is delayed in development. When they become adults, they are short in stature and have a long face. Their jaw is prominent and they usually have protruding ears. Males with Fragile X Syndrome have large testicles, stubby hands and a heart defect. Fragile X Syndrome occurs in about 1 in 1 000 male births and 1 in 2 500 female births.



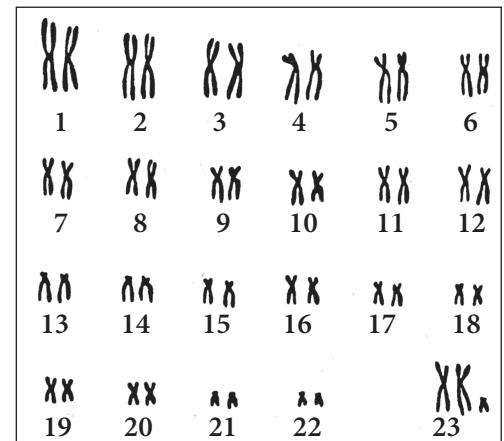
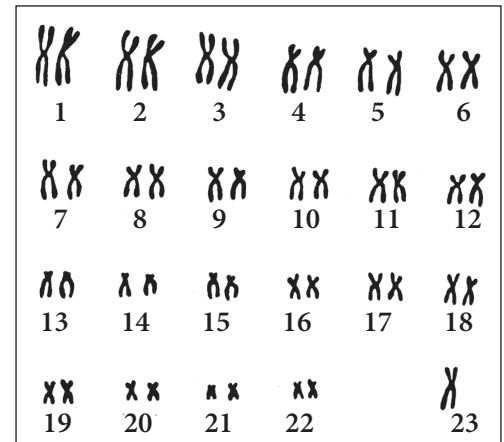


Turner Syndrome

With this condition, too few sex chromosomes have been inherited; the sex chromosomes are written as XO. The O signifies the absence of the second sex chromosome. Females are short and have a broad chest and webbed neck. Their ovaries, oviducts and uterus are very small and non-functional. They do not undergo puberty or menstruate and there is a lack of breast development. They are of normal intelligence and can lead normal lives. The frequency of Turner Syndrome is about 1 in 6 000 births.

Klinefelter Syndrome

These males have inherited two X chromosomes as well as the Y chromosome. As a result, they are sterile. Their testes and prostate glands are underdeveloped and they have no facial hair. They can also have some breast development. These individuals have large hands and feet as well as long arms and legs. They are slow learners but are not mentally retarded. Klinefelter Syndrome happens in about 1 in 1 500 births.



1 For each genetic disease, circle the chromosome responsible in the karyotype.

2 In your workbook, draw up a table like the one below. Complete the table to summarise the information about each inherited genetic disease.

Name	Cause	Symptoms	Frequency

3 Draw lines to match each genetic disease with the affected chromosomes.

- | | | | |
|-------------------------------|------------------------|-------------------------------|--------------------------|
| a Cri du Chat Syndrome | b Down Syndrome | c Klinefelter Syndrome | d Turner Syndrome |
| • deletion in chromosome 5 | • XO | • XXY | • extra chromosome 21 |

Refer to the text below to answer the following questions.

Women over the age of 40 years are warned that they have a greater risk of delivering a baby with Down syndrome. These mothers are given the option to take a test to determine whether their baby has Down syndrome. Then they can make a decision whether to have the baby or not.

Some human rights groups argue that these tests are unethical. They say that every baby has a right to live regardless of whether they have disabilities or not. Other groups believe it is a mother's right to avoid the suffering and hardship with a disabled baby.

4 Explain the meaning of:

- a** unethical. **b** right to live.

5 Explain why suffering and hardship may be experienced when a family has a child with Down Syndrome.

6 Do you think every pregnant woman over 40 years should have to take a Down Syndrome test?

7 a Do you think that parents should be allowed to use these tests to decide the future of their child?

b Do you think that doctors or welfare workers should be allowed to use these tests to decide the future of their patients' child?



Sometimes the codes in genes are changed, and the developing cells receive a wrong message. A change in the codes could cause the cells to make a wrong protein, or no protein at all. As a result, the cells do not function normally. Changes in the genes are called **mutations**, derived from a Latin word meaning *change*. Mutations can be dominant or recessive. If dominant, they will be expressed and hide the normal form of the gene. If recessive, the normal gene will hide the mutation. To have the characteristic of the recessive mutation, a person must have both genes with that mutation.

The table below summarises some mutations or genetic disorders in humans.

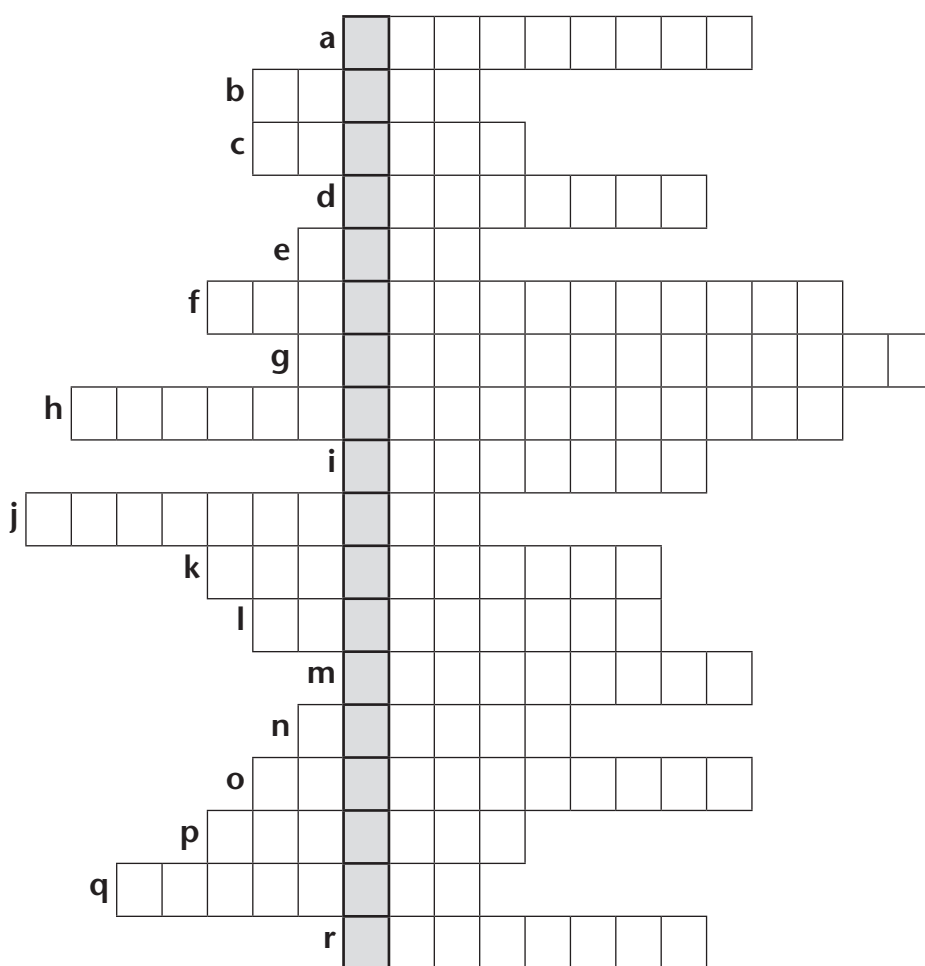
Name of disorder	Description	Frequency	Dominance and position
Albinism	White skin, white hair, pink eyes, poor vision because unable to produce pigment	1 in 20 000	Recessive On eleventh chromosome
Cystic Fibrosis	- Usually lethal - Breathing difficult because mucus in bronchial tubes thick and restricting - Digestion difficult because mucus in pancreatic ducts thick and blocking	1 in 2 500 Caucasians	Recessive On second chromosome
Duchenne Muscular Dystrophy	- Waddling gait, toe walking - Frequent falls, difficulty in getting up - Confined to wheelchair - Death usually at age of about 20 years because of weak and wasted muscles - Males are rarely fathers, but females can be carriers without showing disease but passing on faulty gene	1 in 5 000 males	Recessive Sex linked, on X chromosome
Haemophilia	- Can be lethal - Can bleed to death from minor injury - Internal bleeding particularly around joints - Blood does not clot because clotting agent is absent, so bleeding does not stop	1 in 5 000 males	Recessive Sex linked, on X chromosome
Huntington Disease	- Neurological disorder - Lethal 10 to 15 years after start of symptoms - Normal until middle age, then muscular spasms and personality disorders because of progressive degeneration of brain cells - Affected parents may have affected children before symptoms show	1 in 20 000	Dominant On fourth chromosome
Phenylketouria (PKU)	- Metabolic disorder - Severe mental retardation because enzyme is missing that breaks down amino acid phenylalanine resulting in poisoning cells - Newborns are placed on diet low in phenylalanine until about 7 years, when brain is fully developed	1 in 5 000 caucasians	Recessive On twelfth chromosome
Tourette Syndrome	- A tic disorder: 'twitches and noises' - On average starts at 7 years - Involuntary movements: eye blinking, nose twitching, making grimaces, arm and leg twitches - Involuntary noises: grunting, throat clearing, shouting, saying obscenities, barking - Often only mild symptoms	1 in 1 500 more common in males than females	Dominant On nineteenth chromosome ? Possibly on other chromosomes



- 1** Write a report. Select one genetic disorder and write a report about it. There should be a short introductory paragraph and then several ordered information paragraphs. Use the scaffold below to help you organise your paragraphs. You could also use the library or Internet to research extra information about the disorder.
 - **Introduction:** What is the name of the disorder you have chosen?
 - **Description 1:** Describe the disorder, using complete sentences.
 - **Description 2:** What is the frequency of the disorder? Is it a recessive or dominant disorder? What chromosomes are affected?
 - **Conclusion:** Summarise the disease in one sentence.
- 2** Now select a different genetic disorder. Write a report about it, following the format described in question 1.
- 3** Solve the clues to complete the crossword puzzle. The sentence in the middle of the puzzle is about a person who was a famous musician and who was considered rude and uncontrollable. Geneticists believe he may have suffered from this disease. Ignore the spaces between words in this sentence.

Clues

- | | |
|--|--|
| <p>a Changes in genes are called _____.</p> <p>b Mutations give cells the _____ message.</p> <p>c This is needed to break down the amino acid phenylalanine.</p> <p>d A person who appears pale and has pink-coloured eyes may have _____.</p> <p>e Huntington disease is located on chromosome 4. True or false?</p> <p>f Genetic disorder that causes thick mucus to accumulate in pipes in the lungs and pancreas.</p> <p>g The abbreviation for this disease is PKU.</p> <p>h When children find walking and getting up difficult, they may suffer from _____.</p> <p>i If mutations are expressed with only one gene, they are said to be _____.</p> <p>j A disorder that appears later in life and involves severe muscle spasms.</p> <p>k A sex-linked disease where people lack a blood clotting agent; mainly found in males.</p> <p>l Group of people with a long face, medium-to-tall stature, light-coloured skin, originating from Europe.</p> <p>m People must have two genes for a _____ characteristic to appear.</p> | <p>n Deadly</p> <p>o If newborns have PKU, mental _____ can be avoided if they eat a special diet.</p> <p>p Genes code for _____.</p> <p>q Organ that helps in digestion.</p> <p>r Descriptions of a disease can also be referred to as the _____ of the disease.</p> |
|--|--|





The science of genetics isn't just about the facts and the procedures – geneticists need to think about and answer some tricky ethical dilemmas as well.

Read the scenarios and respond to the questions. Where you are asked to give your opinion, include reasons or an explanation for your response. You may need to refer to other resources to complete some questions.

Scenario one

Michael is an aeroplane pilot. He loves his job. He has been married for seven years and has two children, Olly aged fifteen and Lizzy aged thirteen. Michael lost his mother due to an illness called Huntington's Disease (HD). It was very difficult for Michael to watch his mother deteriorate and die. At first, she had strange mood swings and her arms and legs began to twitch continuously. Soon after, she couldn't control her movements and became bedridden. She could barely eat her food without choking.

Michael knows that symptoms of HD appear in middle age and that there is no treatment. He is also aware that the disease is inherited and that he has a good chance of having the disease gene. Michael has just learnt about a test he could take. The test result will tell him if he carries the disease gene or not.

Michael is tempted to take the test. On one hand, he will find out whether he could get the disease and so plan for the future. On the other hand, he is not sure whether he should tell his employer and his life insurance company.

- a** What does an aeroplane pilot do? Is it a responsible job?
- b** What type of disease is HD?
- c** What causes this disease?
- d** Why won't medication prevent or cure HD?
- e** How could Michael have inherited the disease?
- f** What would the HD test examine?
- g** What forms the HD gene?
- h** How could Michael plan for the future?
- i** Should Michael tell his employer?
- j** Should Michael tell his insurance company?
- k** Is it illegal not to tell employers and insurance companies about positive genetic test results?
- l** If you were Michael, would you take the test?

Scenario two

Kim's father died in a minor car accident when his car ran off a quiet road. He only received a bump to his head which left him with a cut and unconscious. But he simply bled to death, because the blood kept on flowing from the cut on his head.

Kim is pregnant and she is worried. What if her baby bleeds like her father? The baby could die from a small external or internal wound. She thinks she would not be able to bear that. Maybe she should have a test to find out if the baby is a boy, and if it is, have an abortion.



- a** Why did Kim's father bleed to death from a cut to his head?
- b** What is the name of the father's genetic disease?
- c** Which chromosome carries the gene for this disease?
- d** What is the term used to describe Kim if she has that gene but is not a 'bleeder' herself?
- e** Is the 'bleeder' gene dominant or recessive?
- f** How could Kim find out whether the embryo is male or female?
- g** Should Kim have an abortion?
- h** What do you think about abortion? Are there any circumstances where it is justified? Are there any circumstances where it is not justified? Who should be allowed to make this decision?

Scenario three

When Peter and Maria married, they had no idea that they were both carriers of Thalessemia. However, they had a daughter named Millie who has the rare genetic blood disease called Farsces anemia. The disease is fatal. Millie needs a transplant of stem cells with the same tissue type as her own. The only way to accomplish this is to create a 'test tube baby' who would donate healthy cells to Millie. Maria and Peter have asked to choose the sex of the baby, since they had hoped to have a son one day anyway.

If Maria and Peter go ahead with this procedure, in vitro fertilisation will be used to create several embryos. The doctor screens these and chooses a male with the same blood type as Millie. The chosen embryo is then implanted in Maria's uterus. The other embryos will be discarded or frozen. When the baby is born, cells from his umbilical cord will be infused into his sick sister.

The doctor said that parents have a right to give birth to a healthy baby, but she is concerned about the ethics of the procedure. The last thing she wants to see in the media is her photo with the headline: *Designer babies to order: Where do we draw the line?*

- a** How can a person be a carrier of a genetic disease?
- b** What does *fatal disease* mean?
- c** Why are stem cells so important?
- d** Where do the sperm and eggs used for a 'test tube baby' come from?
- e** Is it fair to use the new baby as a donor for Millie?
- f** Should Maria and Peter be allowed to choose the sex of their new child? What else should they choose?
- g** What is the abbreviation for in vitro fertilisation?
- h** Could the new embryo be implanted in the uterus of a different woman? Under what circumstances might this be necessary?
- i** Is the discarding of unused embryos murder?
- j** Should unused embryos be used for medical testing, rather than be discarded?
- k** Do parents have the right to give birth to a healthy baby?
- l** What does *ethics* mean?
- m** What are 'designer babies'? Who should have designer babies?
- n** If you were Maria or Peter, would you have the IVF baby?
- o** Consider the question: *Designer babies to order: Where do we draw the line?*
Where do you think the line should be drawn?

**1** Unscramble the words in brackets to complete the information below.

Genetics is the study of how traits are passed from one (eeaiognrt) _____ to the next. Genetics also looks at why (glnvii) _____ things of the same kind are so similar yet so unique.

People have known for hundreds of years that traits can be (deetnhirit) _____. They observed how appearances were passed on from parent to child, and how illness ran through families, in both people and animals. But no one understood how inheritance really worked. The invention of the (oopirscemc) _____ about 400 years ago allowed scientists to view tiny (lscl) _____ for the first time. They soon discovered that living things were made of cells, and that cells divide to form new cells, which allows living things to (wrgo) _____.

The first studies of inheritance were carried out by an Austrian monk named Mendel in 1866. He studied many (aaihrterstccs) _____ of the garden pea and cross-pollinated one at a time to discover that inheritance followed precise (eulsr) _____.

By 1902, scientist Walter Sutton and biologist Theodore Boveri independently theorised that genes lie on (soommchres) _____. Their hypothesis was based on observations of how chromosomes behaved during (ssioime) _____, a type of cell division. Thomas Morgan went on to show that genes were located on chromosomes which themselves were located in the (suncule) _____ of a cell.

In the 1940s, biologists George Beadle and Edward Tatum made major discoveries while studying the fungus *Neurospora*, a pink mould found on stale bread. They discovered that each gene produced one kind of (troipen) _____. Soon after, biologist Oswald Avery demonstrated that genes are made up of deoxyribonucleic (caid) _____, or DNA.

By the 1950s, scientists were striving to be the first to discover the three-dimensional structure of DNA. Biochemist Maurice Wilkins and scientist Rosalind Franklin worked together using X-ray diffraction crystallography. Franklin produced an image of DNA which showed that the DNA molecule is (eicllah) _____ with more than one (nachi) _____. By 1953, biologists Francis Crick and James Watson had used Franklin's data to discover the three-dimensional (trusteruc) _____ of DNA, which is now known as the double helix.

Gradually scientists began to unravel the mysteries of (snigeetc) _____.

By the 1970s, experiments were conducted where foreign genes were inserted into a (caaribte) _____ to successfully produce foreign protein. The development of modern research tools helped scientists learn more and at a much faster rate. For example, by 1986 Leroy Hood had developed a DNA (ceqseeun) _____. This machine automatically reads the order of the building blocks of DNA.

By 1989, the Human (Gnomee) _____ Project began. The aim of this project was to map the DNA sequence of humans. The project was completed in 2002.

Today, more powerful computers and new discoveries are teaching scientists a great deal about the genetic constructions of the human body and other living things. This new information is providing us with new opportunities to modify living things. However, it also means that people now face new and difficult (licetah) _____ choices.

**2** List at least five ethical choices people now face due to technological developments in the area of genetics.**3** Choose one ethical issue you listed in question 2. Research the issue and find out what the key points are. Decide what *you* think about the issue. Then write a persuasive text to communicate your opinion.**4** Choose one scientist mentioned in the information above. Research the life of this scientist, and develop a presentation explaining how this person has contributed to scientific knowledge.



The first studies of inheritance were carried out by Gregor Mendel in 1866. He studied many characteristics of the garden pea and eventually discovered that inheritance followed precise rules. Mendel chose peas because normally pea plants are self-fertilising and their pollination can be controlled. His research involved careful planning. He used thousands of plants and experimented over eight years. Mendel became the first scientist to understand the importance of statistics in investigations and to apply mathematics to biological problems.

Mendel's first step was to spend two years growing the same peas, ensuring each flower fertilised itself. During fertilisation, pollen from stamens containing the male sperm fuse with a female egg in order to produce a seed. Every time Mendel's offspring plants looked the same as the parent plants. In this way Mendel made sure that the plants were purebred – that is, their chromosomes were homozygous.

1 How do you think Mendel could ensure that each pea flower fertilised itself?

2 Why was it important to make sure that the pea plants were purebred?

3 Use these terms to label the pea flower diagram:
eggs, ovary, petal, pistil, pollen, sepal, stamen.

4 Apply your own maths skills by filling in the missing numbers in the following description of Mendel's method.

Structure of a pea flower



Let's assume that Mendel selected 2 pea types to experiment with: a tall pea plant and a short pea plant. He didn't pay attention to any other characteristics apart from size.

While the flowers of 10 tall pea plants were still in bud, Mendel carefully opened the 2 buds of each plant and cut off the 6 stamens in each one. Altogether he opened 20 buds and removed _____ stamens. He wanted to make sure that the peas would not self-fertilise.

Then Mendel used a small brush to touch the stamens of flowers of other small pea plants, picking up pollen with the brush. He touched the pistils of the 20 flowers of the tall pea plants with the loaded brush. Then each bud was covered with a tiny, labelled bag, which could have said: 'Pollen from short peas'. In a similar manner, Mendel transferred pollen from other tall pea plant flowers to _____ flowers of short pea plants, which had had their stamens removed. Again the buds were bagged and labelled: 'Pollen from tall peas'. Altogether, Mendel wrote _____ labels for the tall and short plants.

Then Mendel waited until seeds had formed. He collected the pods from the _____ little bags and shelled them. If we assume that each pod contained 5 seeds, Mendel ended up with _____ seeds.

When the soil was prepared again, Mendel planted all those seeds. The seeds germinated and grew into plants. Mendel noticed that every one of the new pea plants was tall. Again when 2 buds on each plant had formed (_____ buds in total), he took out the stamens (altogether _____ stamens). But this time he used the pollen from those collected stamens to transfer them to the pistils (altogether 400 pistils). Like before, Mendel bagged and labelled the fertilised buds (altogether 400 labels).

At the end of the season he collected _____ pea pods and shelled them all. If we assume again that each pod contained 5 seeds, Mendel had altogether _____ seeds.

In the next season, (there are 2 pea seasons per year) Mendel planted the seeds. As the pea plants grew, he counted 1 500 tall and 500 small pea plants. He calculated the ratio of tall to small plants by dividing both numbers by the smaller one and arrived at a ratio of _____ tall plants to _____ short plant.



From his experiments, Mendel concluded several things:

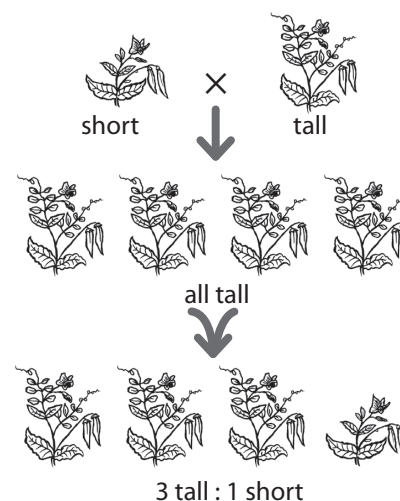
- Each purebred parent (P generation) gave only one factor to its offspring. We now call those factors *genes*.
- Since each plant in the F1 generation (children) had a factor for tallness and a factor for shortness (genotype) but looked tall (phenotype), Mendel determined that tall was dominant over short which was recessive.
- Although each tall factor and each short factor could join with either a tall or a short factor in the F2 generation (grandchildren), most of the F2 generation looked tall.

Mendel was not able to predict exactly what each F2 offspring would look like, but he was able to state that each offspring had a 75% chance for the dominant factor (tall) to appear.

Mendel altogether looked at seven different features – size was only one of them. He learned that each feature had only two expressions; for example, the shape of the pod could be either inflated or wrinkled and the flower colour could be either purple or white. He learned that each feature was independent of every other feature, so it was possible that plants with dark-coloured pods had yellow or green seeds and white or brown seed coats.

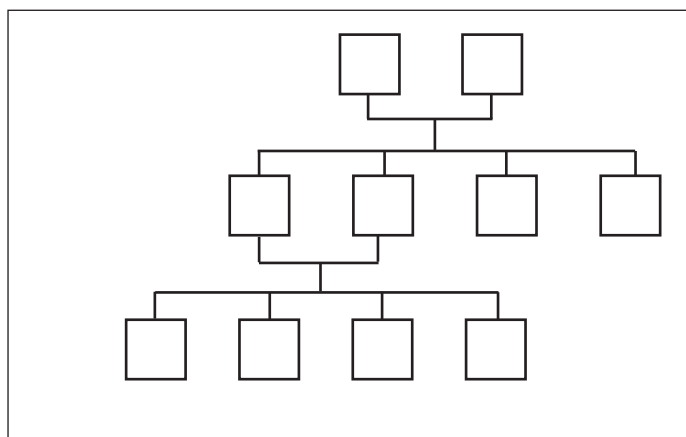
Mendel showed that the appearance of different characteristics in heredity (phenotype) followed two laws. The **law of segregation** states that each feature is determined by two factors (characteristics, now called genes), which are either dominant or recessive, and that only one factor is passed from each parent to the offspring. The dominant factor hides the recessive factor when both factors occur together (genotype); for example, for size tall hides small. The **law of independent assortment** states that factors of traits appear independently of other traits; for example, a tall plant can have either purple or white flowers and round or wrinkled seeds.

Mendel's work was astonishing because he repeated it so many times and applied maths to his process and his results. His findings did not make sense to other scientists of his time, who believed that the parents' characteristics were mixed up in the offspring. It was more than thirty years before scientists recognised that Mendel had laid the foundation for the new discipline of genetics.



- 5** Use the steps below to help you label the family tree for Mendel's peas. For each plant, list the phenotype (tall or short) below the box and its genotype (pairs of genes) inside the box.

- Start with a pure bred tall plant and a pure bred small plant. This is the parent generation (P). The pollen of the small plant are used to pollinate the pistils of the flowers of the tall plant. The flowers develop four seeds which then grow into plants. This is the children generation (F1).
- Only pollen from two F1 plants are used to fertilise the pistils of their flowers. The resulting four seeds again are planted and allowed to grow into plants. This is the grandchildren generation (F2).



- 6** Why did Mendel use pea plants for his experiments, and not dogs or humans?
- 7** Mendel's experiments would have been much quicker if he had used 1 tall plant and 1 short plant, and grown three generations from them. Why didn't he?
- 8** Mendel was not able to predict exactly what each F2 offspring would look like. Why?
- 9** In what ways do you use statistics and other kinds of mathematics in science?
- 10** Using Mendel's method as a guide, describe the steps you think would be involved in the medical testing of a new type of drug. How would statistics and other kinds of maths be used?



The ability to find relevant information, critically analyse it and summarise what it tells you are important to every scientist. Now it's time to test your skills at this process. You need access to the Internet and/or biology and genetics resources to complete this task.

1 Each box below and on the next page contains key words relating to an important discovery about DNA. Use the resources supplied or the Internet to:

- describe what was discovered or achieved
- explain how this contributed to human understanding of DNA.

Record the information described above in note form in each box, and remember to record details of the source of your information (i.e. book title and page number, or website).

1865, Gregor Mendel,
dominant, recessive,
characteristics

1902, Walter Sutton,
chromosomes, cell division

1913, Alfred Sturtevant,
genetic map, chromosomes

1950, Barbara McClintock,
genes, characteristics

1952, Alfred Hershey, Martha
Chase, experiment, genes

1953, James Watson, Francis
Crick, Maurice Wilkins,
Rosalind Franklin, chemical
structure

1956, Jo Hin Tjio, Albert
Levan, 46, human

1958, Matthew Meselson,
Franklin Stahl, DNA,
replicated

1967, Har Gobind Khorana,
Marshall Nirenberg, amino
acids



1970, restriction enzymes,
DNA

1977, nucleotides, DNA
molecule, virus, Fred Sanger,
Walter Gilbert, Allan Maxam

1978, Louise Brown, IVF

1978, Genentech, gene,
insulin

1984, Alec Jeffries,
DNA sequence

1986, GM crops,
herbicide-resistant

1989, human gene, Francis
Collins, Lap-Chee Tsui, cystic
fibrosis.

1997, Hunt Willard, artificial,
chromosome

1997, Roslyn Institute, Dolly,
sheep

1998, Dolly, Bonnie

2003, The Human Genome Project

2 Now use a word-processing program or a large sheet of poster paper to create a timeline including the events listed above. Also include any other events you think are significant to the development of our understanding of DNA. Where possible, add labelled diagrams or illustrations relevant to each event.

**1: Meiosis – One cell becomes four sex cells**

- 1** **a** chromosome **b** Nuclear, homologous **c** separated, spindle **d** apart, cell **e** cell, pair. Stage 1 [in order from left to right] b, e, d, c, a
- 2** **a** division, parent **b** split (or separate) **c** spindle, membrane **d** disappear, separate (or split) **e** cell, chromosome. Stage 2 [in order from left to right] c, e, b, d, a
- 3** Stage 1 finishes with two cells, stage 2 with 4. Stage 1 has a full complement of chromosomes, stage 2 has half. Duplication of chromosomes occurs in stage 1 not stage 2. Chromatids split in stage 2; in stage 1 homologous pairs separate.
- 4** **a** Cell division that results in four cells, only occurs in sex cells, halving the number of chromosomes in a cell **b** Pair of chromosomes similar in size and genes **c** The centre of a cell **d** Separate ends of a cell **e** Half the number of chromosomes in a cell; i.e. one set **f** Complete set of chromosome number in a cell; i.e two sets
- 5** Too many chromosomes in the nucleus. Each generation will have a different number of chromosomes.
- 6** Produces new combinations in chromosomes, therefore more variation in offspring.
- 7** **a** 4 **b** No **c** Increase variation
- 8** **a** Yes **b** Crossing over **c** Made the way meiosis and crossing over work clearer. **d** Cannot see the cells dividing or pinching in to form the new cells.

2: Mitosis, meiosis – what's the difference

- 1** [In order from top to bottom] a, e, b, f, g, c, d
- 2** Sample answer: Explaining mitosis
Mitosis is a type of cell division. During mitosis, cells and their contents divide to produce identical daughter cells.
Just before mitosis of a cell is ready to start, the hair-like chromosomes inside the nucleus double and then coil. As a result, the chromosomes become shorter and visible. When the nuclear membrane dissolves, the doubled chromosomes line up along the middle of the cell. At the same time, the spindle begins to grow from the ends of the cell. As soon as the spindle is complete, it pulls on the chromosomes causing them to separate. Therefore the chromosomes move towards the ends of the cell. After the chromosomes congregate near the ends of the cell, the spindle dissolves. Furthermore, the cell starts to divide. Then the chromosomes uncoil thus becoming invisible. The nuclear membrane also forms around the chromosomes. Finally, the cell separates into two daughter cells.
Mitosis has three main purposes: it occurs when organisms grow bigger, when organisms replace lost or damaged cells, and when simple organisms reproduce asexually.
- 3** [In order from left to right then top to bottom] j, e, a, k, f, h, c, b, g, d, i
- 4** Sample answer:
a A cell contains a membrane, cytoplasm and a nucleus.
b Two pairs of doubled chromosomes are visible in the nucleus. **c** The nuclear membrane has disappeared, the chromosome doubles are lined up in the middle of the cell and the spindle has formed. **d** The spindle pulls the chromosome doubles apart and towards the ends of the cell. **e** The chromosomes congregate at the ends of the

cell, the spindle has disappeared. **f** The cell has divided into two new cells and the nuclear membrane forms around the chromosomes in each cell. **g** The chromosomes uncoil and become invisible. **h** In each new cell the nuclear membrane disappears, the chromosomes line up in the middle of the cell and the spindle attaches to them. **i** The spindle pulls the chromosomes to the ends of each cell. **j** The cells start to split and the chromosomes are at each end without the spindles. **k** The two cells have divided forming four cells. The two chromosomes in each cell have uncoiled and become invisible. The nuclear membranes have formed around the chromosomes.

- 6** See table at right.

Mitosis	Meiosis
All cells	Sex organs
1	2
2	4
46	23
yes	no
yes	no

3: Duplicating DNA

- 1** Hexagonal = sugar, circle = phosphate, base A = one-point rectangle, base T = two-point rectangle, base C = curve out rectangle, base G = curve in rectangle.
- 3** No differences
- 4** No differences
- 5** Each base has only one match.
- 7** In the top strand, a base is folded away and so is not available for the new strand; the fourth base in the low strand is not matching the fourth base of the top strand.
- 8** Death of the cell.
- 9** Change in order of bases resulting in new characteristics.
- 10** X-rays are radiation and may cause mutations in the foetus.
- 11** It could cause mutations in the foetus, resulting in illness, deformity or death.

4: Amazing DNA – reading the DNA code

- 1** egg, zygote, organic, things, discovery, significant, nucleotides, strand, helix, important, together, code, differs
- 3** Deoxyribose sugar in DNA, ribose sugar in RNA; DNA double stranded, RNA single stranded; DNA differs by one base – has thymine instead of uracil in RNA.
- 4** A small unit of DNA made up of a sugar, phosphate and nitrogenous base.
- 5** Could change an amino acid sequence therefore resulting in a different protein.

Complete name	Codes
Valine	GUU, GUA, GUG
Tryptophan	UGG
Threonine	ACU, ACC, ACA, ACG
Serine	AGU, AGC, UCU, UCC, UCA, UCG
Proline	CCU, CCC, CCA, CCG
Phenylalanine	UUU, UUC
Methionine	AUG
Leucine	CUU, CUC, CUA, CUG, UUA, UUG
Isoleucine	AUU, AUC, AUA
Glutamine	CAA, CAG
Cysteine	UGU, UGC
Aspartic acid	GAU, GAC

- 7** Person 1:
m-RNA code: AAG CCC UAC GUG UUU UCC AUC
Protein: Lys – Pro – Tyr – Val – Phe – Ser – Ile



Eye colour: Dark brown

Person 2:

m-RNA code: AAG GUA UAC CCG UCU UUG AUU

Protein: Lys – Val – Tyr – Pro – Ser – Leu – Ile

Eye colour: Hazel

Person 3:

DNA code: TTC GGA ATG GGC AAA AGG TAG

Protein: Lys – Pro – Tyr – Pro – Phe – Ser – Ile

Eye colour: Light brown

Person 4:

DNA code: TTC CAT ATG AGA GGC AAC TAA

Protein: Lys – Val – Tyr – Ser – Pro – Leu – Ile

Eye colour: Blue

8 a AAG GUA UAC UCU CCG UUG AUU

b Lys – Val – Tyr – Ser – Pro – Leu – Ile **c** Blue

5: Let's make protein

1d, 2b, 3e, 4a, 5f, 6c, 7h, 8g

Sample answer: Making protein

DNA sends a message to ribosomes in the cytoplasm to make a particular protein.

The DNA strands unzip and a copy of a section of DNA code is made by matching its bases with complementary bases. These complementary bases form m-RNA. When the m-RNA is complete, it moves out of the nucleus into the cytoplasm and then attaches itself to a ribosome. t-RNA attaches itself to the m-RNA – every three bases of the t-RNA should match three bases of the m-RNA. Because each three bases of t-RNA carry one specific amino acid, the m-RNA determines the order of the amino acids. When the t-RNA meets the stop codon on the m-RNA, the joining of amino acids is finished, resulting in a protein. This protein leaves the ribosome and is used by the cell.

Proteins are very important for the cell – they help the organism to grow and repair itself.

6: Modelling protein synthesis

1 Green, blue, blue, red, red, yellow, red, blue, yellow

2 Yellow, red, red, white, white, green, white, red, green

5 CTTAAGATG

6 GAAUUCUAC

7 CUU AAG AUG

8 For example, the difference between transcription and translation, how bases pair up, that DNA has base thymine and RNA has uracil, how proteins are assembled.

9 Nucleus

10 DNA unzips and is copied by m-RNA in the nucleus. Once m-RNA is assembled, it moves out of the nucleus.

11 Cytoplasm on a ribosome

12 m-RNA attaches to ribosome, t-RNA attaches to complementary codes on m-RNA, amino acids join together by peptide bonds to form protein.

13 a UAGCCGUAUAGAUGA

b AUC GGC AUA UCU ACU

15 b For example, no enzymes catalysing the reactions all the way; how ribosome aids in amino acids assembling not shown.

c For example, beads instead of paper for amino acids.

7: Cloning animals – model discussion

1 Introduction – paragraph 1; Arguments for – paragraphs 2, 3; Arguments against – paragraphs 4, 5; Conclusion – paragraph 6

2 The Tasmanian tiger, *thylacine*; Thylacines were native animals of Tasmania; Cloning technology for animals exists; the Tasmanian tiger lived in Tasmania more than one hundred years ago; many cells with a new nucleus never develop into offspring; the Tasmanian tiger should not be cloned.

3 Historical information about Tasmanian tiger; reintroducing the Tasmanian tiger; animals can be cloned; Tasmania has changed; many clones die; don't clone the Tasmanian tiger

4 To decide whether to clone Tasmanian tigers and to convince readers of your conclusion.

5 e.g. thylacine, marsupial, carnivore, native animal, exterminated

6 e.g. only, feared, still, right, wrong, thoughtless, tragic

9: What does genetics have to do with me?

1 Tongue rolling: 73%, 27%

Attached earlobes: 60%, 40%

Receding hair on the sides of the forehead: 50%, 50%

Gap between front teeth: 20%, 80%

Right thumb on top during hand folding: 80%, 20%

Hair on the second digits of your fingers: 60%, 40%

Right dominant hand: 80%, 20%

Second toe longer than big toe: 50%, 50%

Straight hair: 67%, 33%

3 Some characteristics have very uneven occurrences (e.g. tongue rolling, front teeth gap), while others are more equal (e.g. receding hair, longer second toe).

5 Some can be influenced by the environment, e.g. the gap in the front teeth can be closed by braces, hair on your fingers can be waxed off, practice can improve writing with your right hand.

6 Sample answer: I used to have a gap between my front teeth (inheritance), but I wore braces for a long time and now the gap is closed (environment). When I started school, I wrote with my left hand (inheritance), but I learnt to write with my right hand (environment). I had dark hair (inheritance), but now I colour my hair (environment). So I think that my development has been influenced more by the environment.

10: Is it a boy or a girl?

1 In every generation the number of chromosomes would double, and the cells could not function.

2 Diploid – two full sets of chromosomes, 46; haploid – half the number of chromosomes, 23, only one set.

3 a The first 22 pairs of chromosomes, which hold most of the genes. **b** The arrangement of the full set of 46 chromosomes in pairs, from largest to smallest

4 a 46 **b** 46 **c** 23 **d** 23 **e** 46

5 The union of a female X chromosome with a male X chromosome at fertilisation results in a baby girl, whereas the union of a female X chromosome with a male Y chromosome at fertilisation results in a baby boy.



- 6** Body cells, also known as somatic cells, undergo a cell division known as mitosis. These cells have a complete set of chromosomes known as diploid. The purpose of mitosis in these cells is for growth and repair. Sex organs contain sex cells like sperm or ovum. These cells undergo a special cell division known as meiosis. This cell division halves the number of chromosomes from 46 (diploid) to 23 (haploid). The purpose of this cell division is to create the gametes (sex cells).
- 7** **a** Male, because the chromosomes in the 23rd pair are different in size, therefore must be chromosomes XY
b 46 **c** 23 **d** 23 **e** 46 **f** 23 **g** 23 **h** 46
- 8** **a** Males determine the sex of the baby because they produce two different kind of sperm, X and Y. The chances of a female X egg and male Y sperm uniting is 50%. **b** Nothing about inheritance, chromosomes and genes was known then. Technology like the microscope did not exist.

11: Rabbits reproducing

- 2** **a** 6 **b** Yes **c** 3 pairs **d** Dark is dominant, light is recessive.
- 3** **a** Sperm, eggs (ova) **b** 3 **c** Yes **d** Yes **e** 50 **f** 50
- 5** No
- 6** e.g. 80% and 20%

12: Random rabbits

- 1** e.g. 30, 60, 10
- 2** e.g. $30 \div 10$, $60 \div 10$, $10 \div 10$, ratio is 3 : 6 : 1.
- 7** **a** Dark = D; light = d **b** DD, Dd; Dd, dd
- 8** 1 : 2 : 1
- 9** 3 : 1
- 10** Breed more rabbits, say 50 pairs.
- 11** Dark
- 12** **a** Dark **b** Dark **c** Dark
- 13** **a** 150 **b** 50 **c** 50 **d** 100
- 14** No, they are the same colour.
- 15** No
- 16** Yes
- 17** No

13: Genetics with the Beeple people

All students will have different beeeple babies.

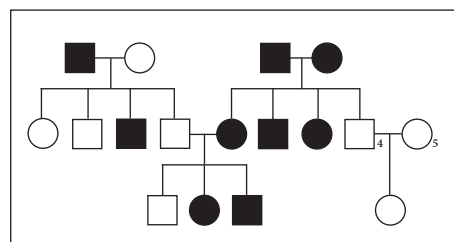
- 2** Mum: XX, BB, Rr, Ss, Ll, pp, aa, Nn
Dad: XY, bb, Rr, ss, ll, Pp, Aa, NN
- 7** **a** 16 **b** 16
- 11** $2 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 4\,374$

14: Where do I fit in my family tree?

- 1** **a** 7 **b** 5 **c** 3 **d** No **e** brother–sister **f** uncle–niece
g maternal grandparents **h** Shade all circles.

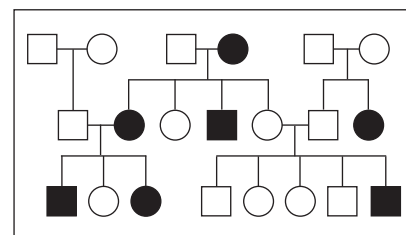
15: Pedigree barrier game

Tongue rolling family pedigree



- 1** **a** 8 **b** 9 **c** 4 **d** 4 **e** 2
- 2** The child does not have the gene for tongue rolling.
- 3** Yes, most family members can roll their tongue and no generation is skipped.

Pentadactyly complete family pedigree



- 4** **a** 8 **b** 3 **c** 8 **d** 2
- 5** Recessive, because a married couple without the characteristic has a child with the characteristic. This means they must both carry the gene and they passed it on to their child.
- 6** 3
- 7**
- 8** See pedigree above right. This is recessive because Mary's mother and father do not have attached ear lobes, but they must carry the recessive gene because Mary has attached earlobes. Mary's two recessive genes are one from her mother and one from her father.

16: The language of genetics

- 1** Nouns: e.g. colour, daughter, son, breed, origin.
Verbs: e.g. to rule, to produce, to show.
- 2** **a** Other shape = matching genes, pair of genes **b** Coloured body **c** Hollow form = material inside cell **d** Ruled = strong, covers up **e** Origin = something that determines characteristics **f** Produced sons and daughters = children generation **g** Produce model = genes which determine appearance, e.g. Bb **h** Spiral **i** Different genes in pair, i.e. Bb **j** Same genes in pair, i.e. BB **k** Mongrel = mixed genes in pair, heterozygous alleles **l** Kernel = smallest building block of DNA **m** Kernel = control centre **n** Produced mother and father = parent generation **o** Show model = appearance, looks, e.g. brown hair **p** Retreated = covered up by dominant gene **q** Joined = first cell of offspring when egg and sperm have joined
- 4** Homozygous
- 5** Genotype
- 6** Tall body size, short body size



- 7 **a** Black **b** Bb (black is dominant) **c** Black fur, B **d** White fur, b **e** White and black fur, Bb **f** Yes **g** Heterozygous
- 8 nucleus, cytoplasm, chromosomes, helix, nucleotides, nucleotides, genes, chromosome

17: Genetic diseases – chromosomal

- 3 **a** Deletion in chromosome 5 **b** Extra chromosome 21
c XXY **d** XO

18: Genetic diseases – mutations

- 3 **a** mutations **b** wrong **c** enzyme **d** albinism **e** true
f cystic fibrosis **g** phenylketouria **h** muscular dystrophy
i dominant **j** Huntington **k** hemophilia **l** caucasian
m recessive **n** lethal **o** retardation **p** protein
q pancreas **r** symptoms

19: Scenarios – what do you think?

Scenario one

- b** Genetic disease **c** An inherited disease **d** Medication can't change the genes. **e** From his mother who passed on the gene **f** Chromosomes and genes to see if the gene is faulty **g** A mutation

Scenario two

- a** His blood did not clot **b** Haemophilia or bleeder's disease **c** X chromosome **d** She is a carrier for that disease.
e Recessive **f** Ultrasound or genetic testing

Scenario three

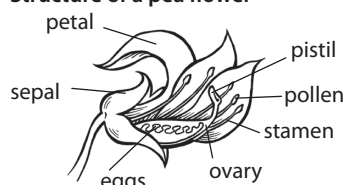
- a** The person has one disease gene which is recessive.
b The disease will cause death. **c** Stem cells can develop into organs and babies. **d** Eggs from the mother's ovary (operation), sperm from the father's testicles (ejaculation)
g IVF **h** Yes, if the mother cannot carry a foetus **i** Deals with right and wrong **m** Offspring with certain features

20: History of the science of genetics

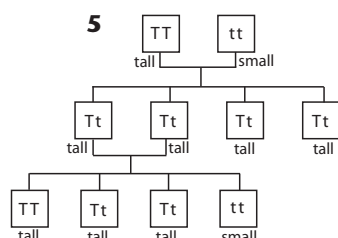
- 1 generation, living, inherited, microscope, cells, grow, characteristics, rules, chromosomes, meiosis, nucleus, protein, acid, helical, chain, structure, genetics, bacteria, sequence, Genome, ethical

21: Mendel's scientific method

- 1 Cover each flower with a bag, so other pollen could not fertilise it.
2 So that Mendel knew the exact characteristics.
3 **Structure of a pea flower**



- 4 120, 20, 40, 40,
200, 400, 2400,
400, 2000, 3, 1



- 6 Because normally peas are self-fertilising, their pollination can be controlled and there are two generations each year.
- 7 Experiments are more accurate the more test subjects there are, and this would not allow him to ensure that there were no recessive genes in the plants he was testing.
- 8 Because the ratios only provide probabilities, not predictions for specific plants.

22: DNA – Find out for yourself

- 1865: Gregor Mendel (Austria) reports that characteristics in peas are determined by two factors: a dominant and a recessive factor; and that characteristics in offspring are predictable.
- 1902: Walter Sutton (USA) shows that chromosomes inside the nucleus occur in pairs and separate during cell division.
- 1913: Alfred Sturtevant (USA) makes the first genetic map of a chromosome, showing where genes are on chromosomes.
- 1950: Barbara McClintock (USA) discovers that genes turn on or off certain characteristics.
- 1952: Alfred Hershey and Martha Chase (both USA) show experimentally that genes are on DNA molecules.
- 1953: James Watson (USA), Francis Crick (UK), Maurice Wilkins (UK), Rosalind Franklin (UK) discover the chemical structure of DNA.
- 1956: Jo Hin Tjio (China) and Albert Levan (Sweden) show that humans have 46 chromosomes in each body cell.
- 1958: Matthew Meselson and Franklin Stahl (both USA) demonstrate that DNA is replicated when the double helix splits open and accepts matching nucleotides to produce two new identical DNA molecules.
- 1967: Har Gobind Khorana (India) and Marshall Nirenberg (USA) determine the DNA sequence for the 20 most common amino acids.
- 1970: Restriction enzymes are discovered, which enables scientists to cut and paste DNA.
- 1977: The order of nucleotides in a whole DNA molecule is found for the first time by Fred Sanger (UK), Walter Gilbert and Allan Maxam (both USA), all working independently. Sanger describes the entire genome for a virus.
- 1978: Louise Brown, the first test tube baby, is born in England.
- 1978: USA company Genentech clones the gene for insulin; i.e. they make a copy of the gene artificially.
- 1984: Alec Jeffries (UK) develops DNA fingerprinting by using the unique DNA sequence in every person.
- 1986: Field trials of genetically modified (GM) crops begin around the world. Many are herbicide-resistant.
- 1989: The first human gene is sequenced by Francis Collins (USA) and Lap-Chee Tsui (China).
- 1997: Hunt Willard (USA) creates the first artificial human chromosome, making designer babies a possibility.
- 1997: At the Roslyn Institute in Scotland, the sheep Dolly is cloned.
- 1998: Dolly gives birth to Bonnie, a healthy lamb conceived by conventional means.
- 2003: The Human Genome Project is completed with 99% of the 1 billion nucleotides and 25 000 genes sequenced.

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